

Module -1 – Introduction and MOS Transistor

By

Dr.Rohit Lorenzo

Associate Professor

School of Electronics Engineering (SENSE)



Introduction to IC Technology

- ▶ There is no doubt that our daily lives are significantly affected by electronic engineering technology.
- ▶ We know revolutionary changes have taken place in a relatively short time and it is also certain that even more-dramatic advances will be made in the next decade.
- ▶ Electronics as we know it today is characterized by reliability, low power dissipation, extremely low weight and volume, and low cost, coupled with an ability to cope easily with a high degree of sophistication and complexity.
- ▶ Electronics, and in particular the integrated circuit, has made possible the design of powerful and flexible processors which provide highly intelligent and adaptable devices for the user.
- ▶ Integrated circuit memories have provided the essential elements to complement these processors and, together with a wide range of logic and analog integrated circuitry, they have provided the system designer with components of considerable capability and extensive application

Micro-electronics

Evaluation

- ▶ The invention of the transistor by William B. Shockley, Walter H. Brattain and John Bardeen of Bell Telephone Laboratories was followed by the development of the Integrated Circuit (IC).
- ▶ The very first IC emerged at the beginning of 1960 and since that time there have already been four generations of ICs:
- ▶ **SSI** (small scale integration), **MSI** (medium scale integration), **LSI** (large scale integration), and **VLSI** (very large-scale integration).

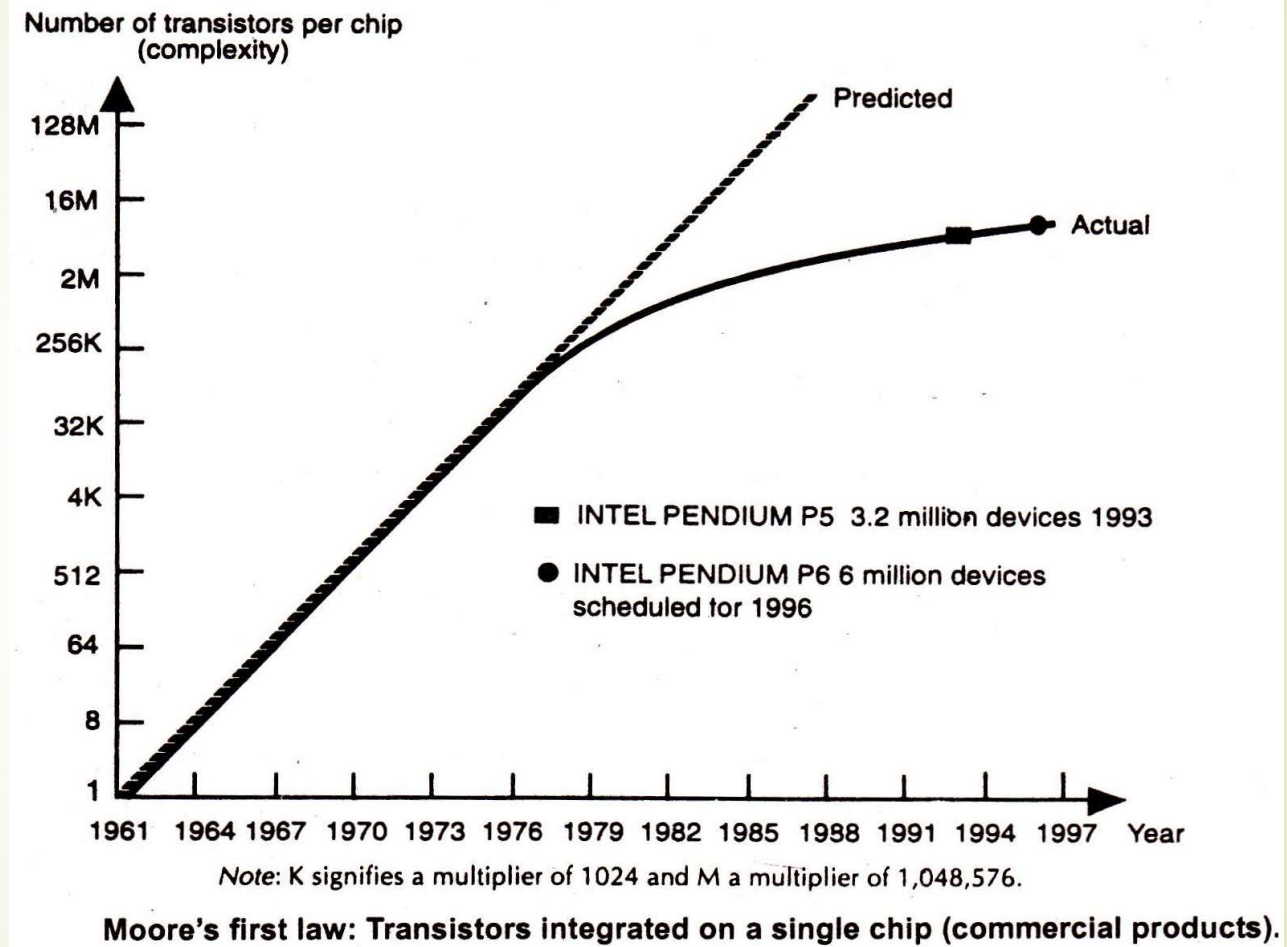
Year	1947	1950	1961	1966	1971	1980	1990	2000
Technology	Invention of the transistor	Discrete components	SSI	MSI	LSI	VLSI	ULSI*	GS†
Approximate numbers of transistors per chip in commercial products	1	1	10	100–1000	1000–20,000	20,000–1,000,000	1,000,000–10,000,000	>10,000,000
Typical products	—	Junction Transistor and diode	Planar devices Logic gates Flip-flops	Counters Multiplexers Adders	8 bit micro-processors ROM RAM	16 and 32 bit micro-processors Sophisticated peripherals GHM Dram	Special processors, Virtual reality machines, smart sensors	

* Ultra large-scale integration

† Giant-scale integration

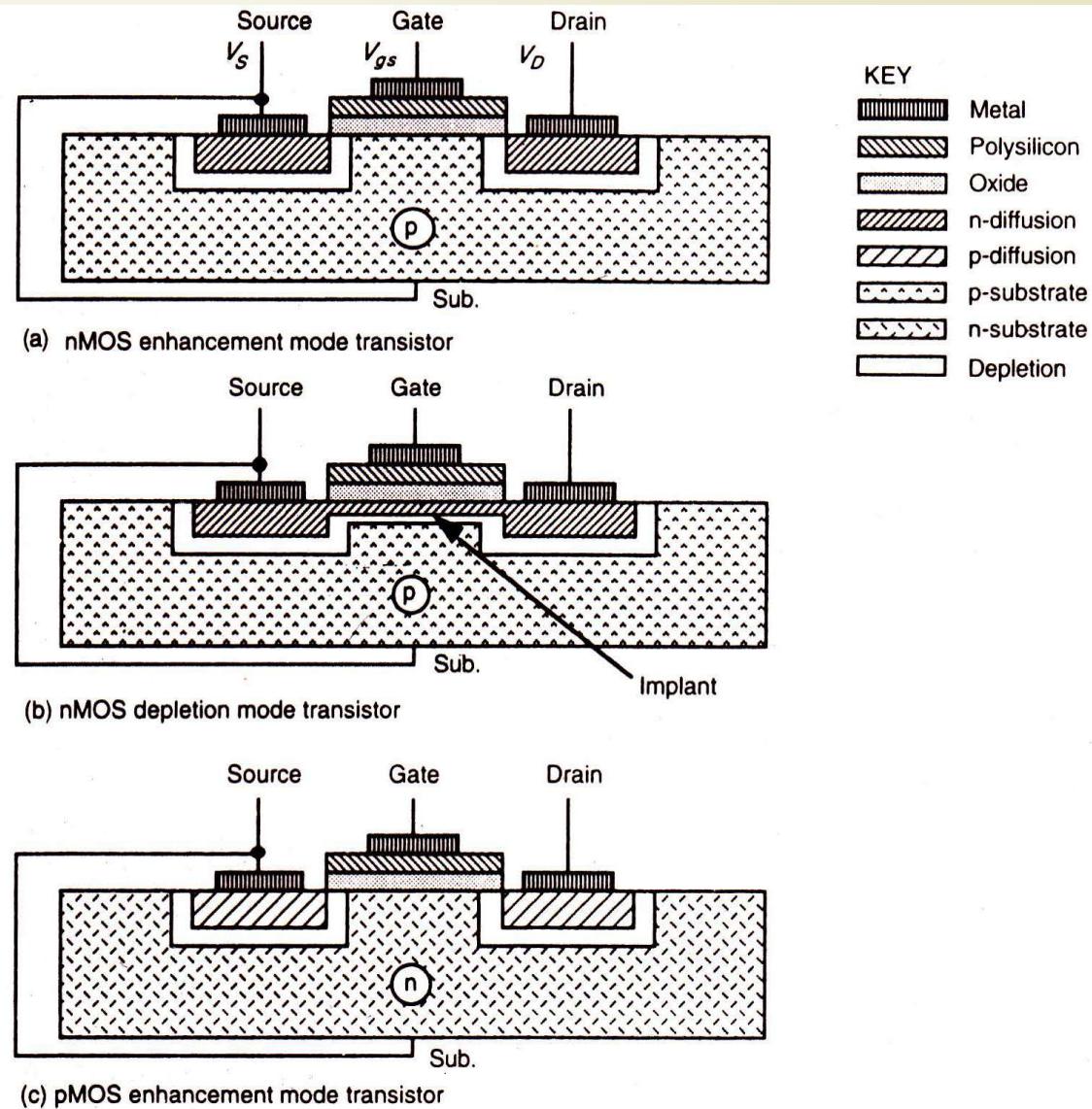
Moore's Law

- ▶ **Moore's law** - prediction made by American engineer Gordon Moore in 1965 that the number of transistors per silicon chip doubles every year.
- ▶ **Moore's law** - Moore observed that the number of transistors on a computer chip was doubling about every 18–24 months.



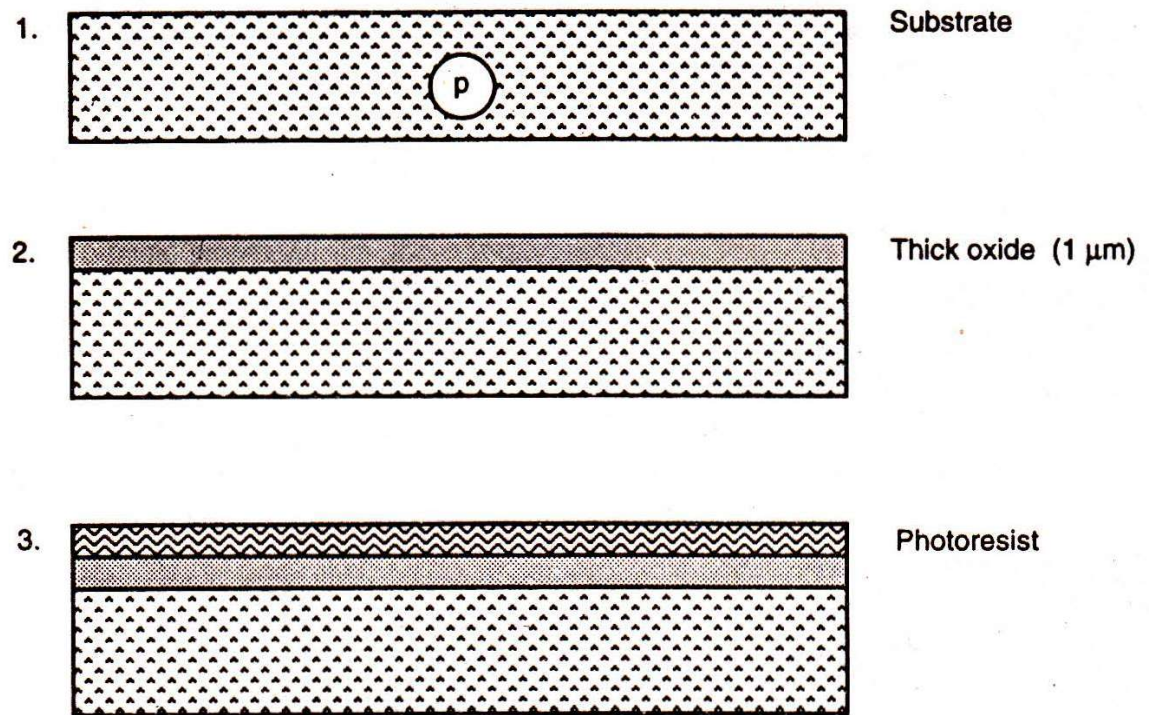
Basic MOS Transistors

- In particular, let us examine the basic nMOS enhancement and depletion mode transistors.



nMOS Fabrication

1. Processing is carried out on a thin wafer cut from a single crystal of silicon of high purity into which the required p-impurities are introduced as the crystal is grown.
2. A layer of silicon dioxide (SiO_2), is grown all over the surface of the wafer to protect the surface, act as a barrier to dopants during processing, and provide a generally insulating substrate on to which other layers may be deposited and patterned.
3. The surface is now covered with a photoresist which is deposited onto the wafer and spun to achieve an even distribution of the required thickness.

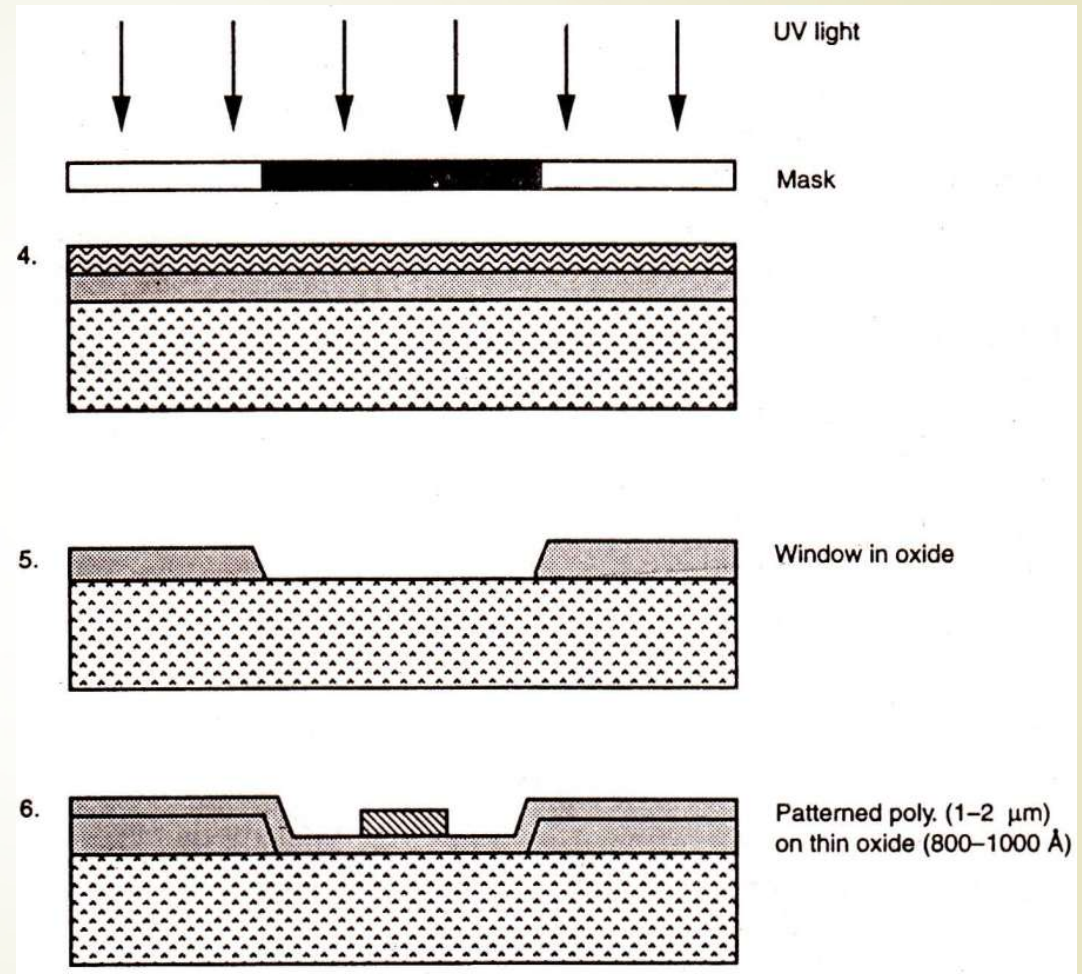


Continue.....

4. The photoresist layer is then exposed to ultraviolet light through a mask which defines those regions into which diffusion is to take place together with transistor channels. Assume, for example, that those areas exposed to ultraviolet radiation are polymerized (hardened), but that the areas required for diffusion are shielded by the mask and remain unaffected.

5. These areas are subsequently readily etched away together with the underlying silicon dioxide so that the wafer surface is exposed in the window defined by the mask.

6. The remaining photoresist is removed and a thin layer of SiO_2 is grown over the entire chip surface and then polysilicon is deposited on top of this to form the gate structure.

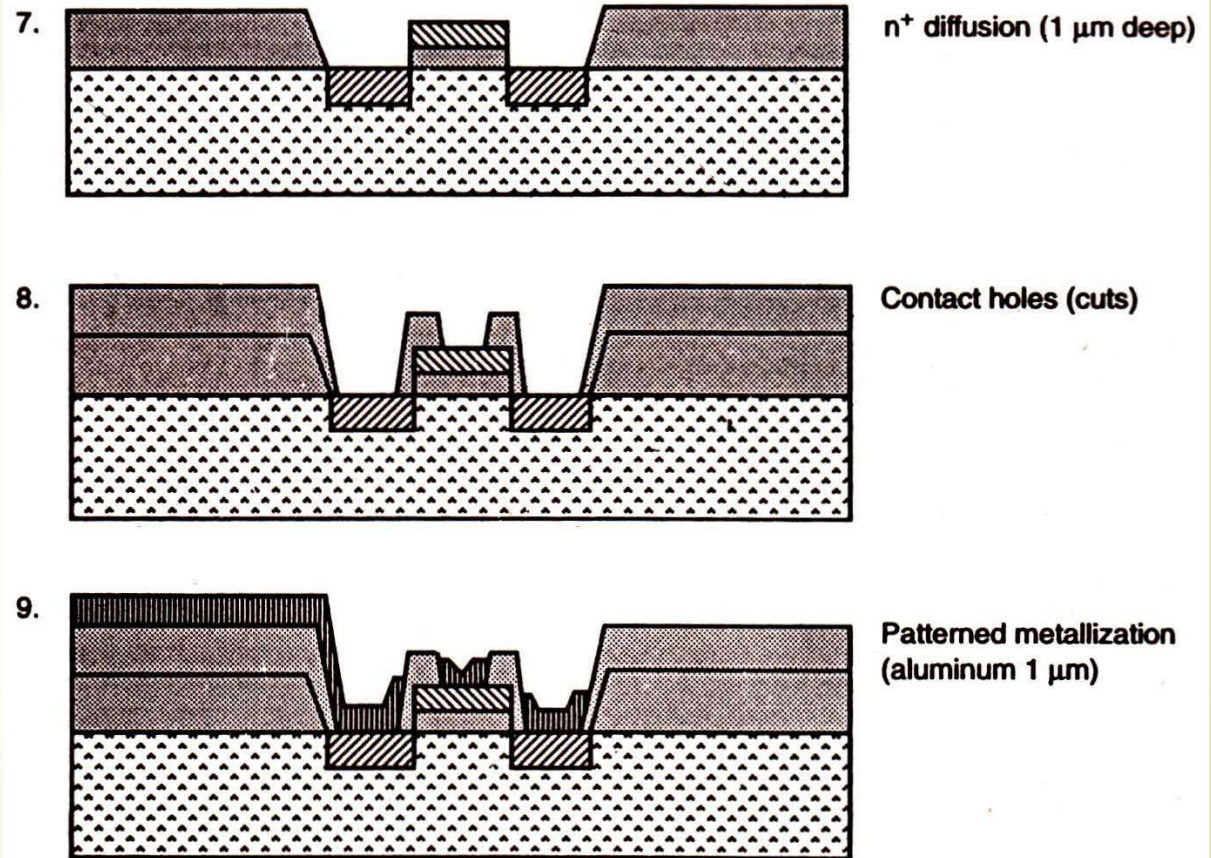


Continue.....

7. Further photoresist coating and masking allows the polysilicon to be patterned and then the thin oxide is removed to expose areas into which n-type impurities are to be diffused to form the source and drain as shown.

8. Thick oxide (SiO_2) is grown over all again and is then masked with photoresist and etched to expose selected areas of the polysilicon gate and the drain and source areas where connections (i.e. contact cuts) are to be made.

9. The whole chip then has metal (aluminum) deposited over its surface. This metal layer is then masked and etched to form the required interconnection pattern.



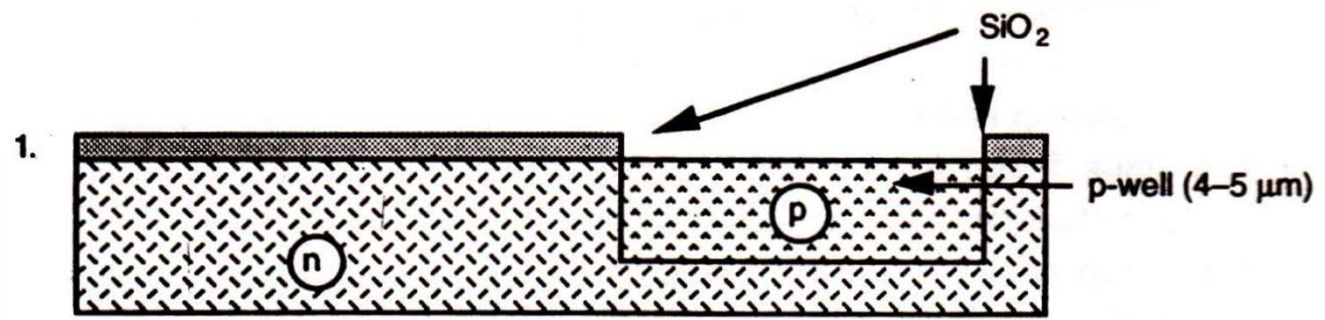


Continue...

- It will be seen that the process revolves around the formation or deposition and patterning of three layers, separated by silicon dioxide insulation. The layers are diffusion within the substrate, polysilicon on oxide on the substrate, and metal insulated again by oxide.

CMOS Fabrication

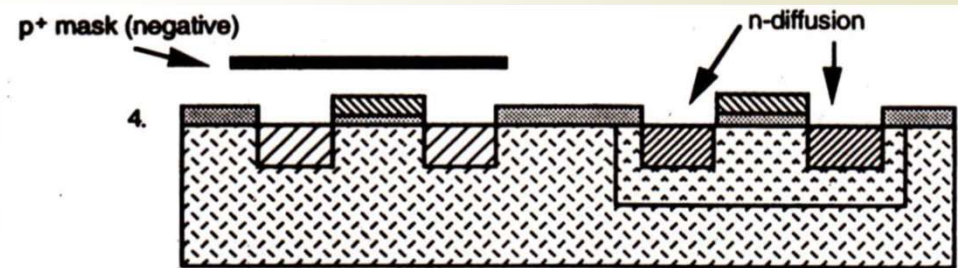
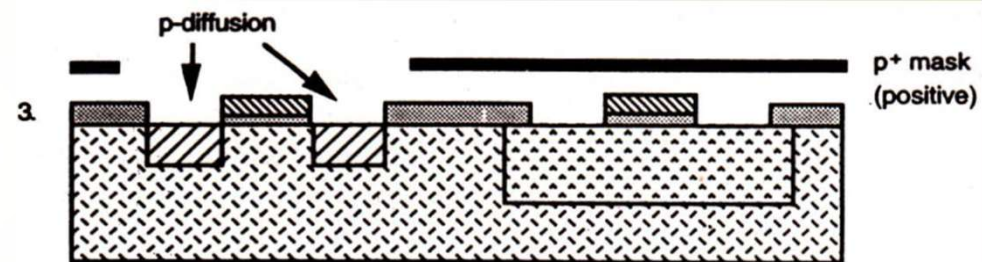
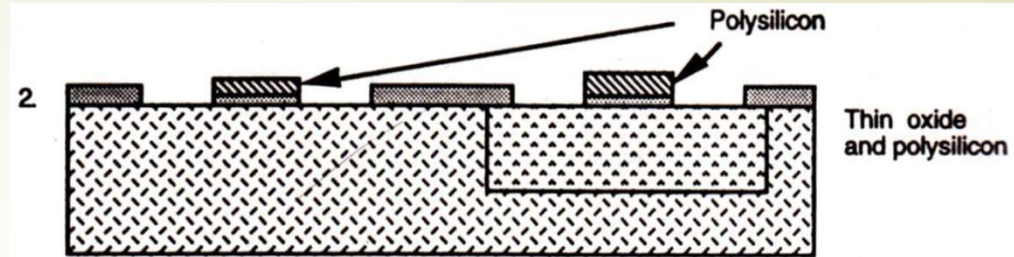
- There are a number of approaches to CMOS fabrication, including the p-well, the n-well, the twin-tub, and the silicon-on-insulator processes.
- The p-well process is widely used in practice and the n-well process is also popular. For the lambda-based rules set out later, we will assume a p-well process.
- **P-Well Process**
- In all other respects—masking, patterning, and diffusion—the process is similar to nMOS fabrication. In summary, typical processing steps are:
- *Mask 1* - defines the areas in which the deep p-well diffusions are to take place.



Continue....

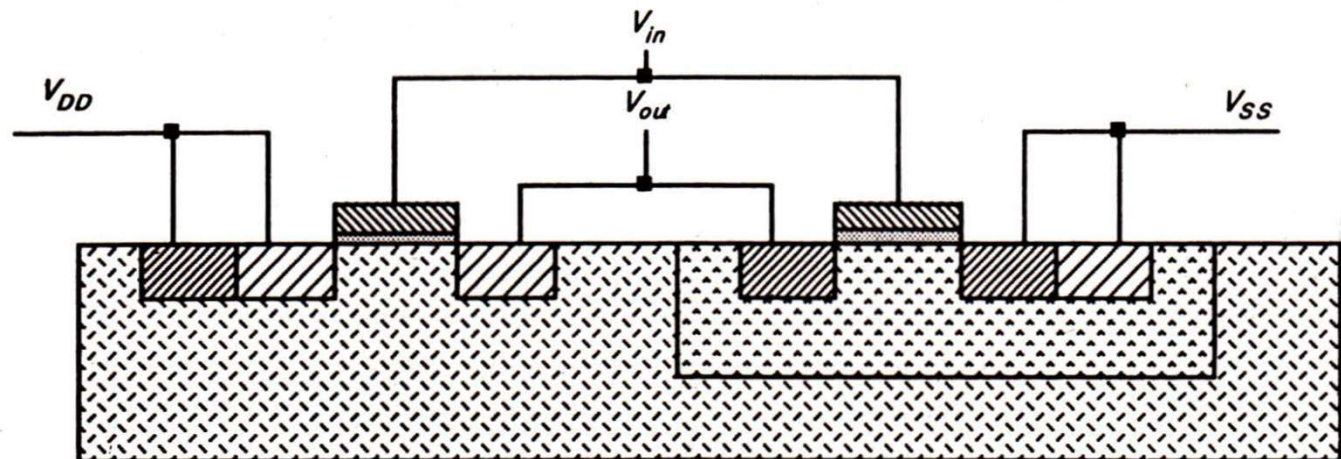
- *Mask 2* - defines the thinox regions, namely those areas where the thick oxide is to be stripped and thin oxide grown to accommodate p- and n-transistors and wires.
- *Mask 3* - used to pattern the polysilicon layer which is deposited after the thin oxide.
- *Mask 4* - A p-plus mask is now used (to be in effect "anded" with Mask 2) to define all areas where p-diffusion is to take place.

Rohit Lorenzo, VIT-AP

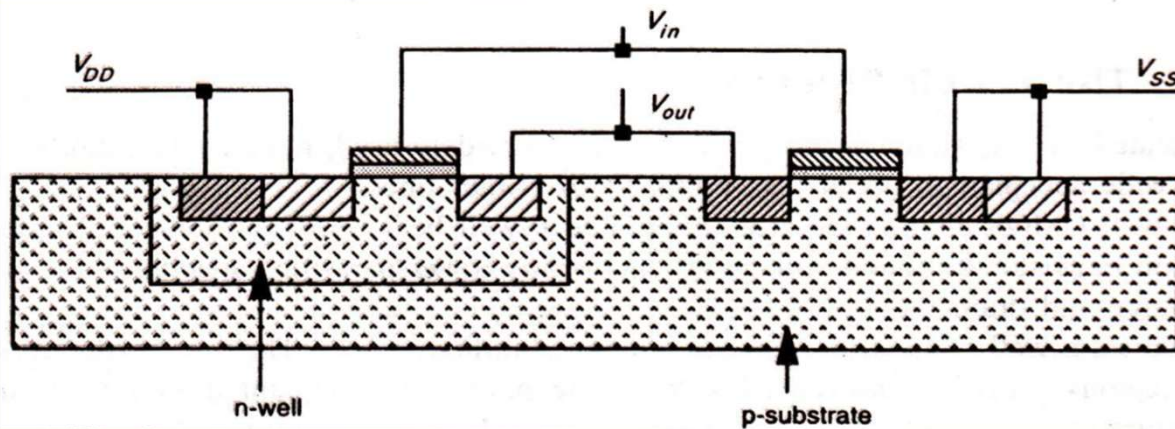


Continue.....

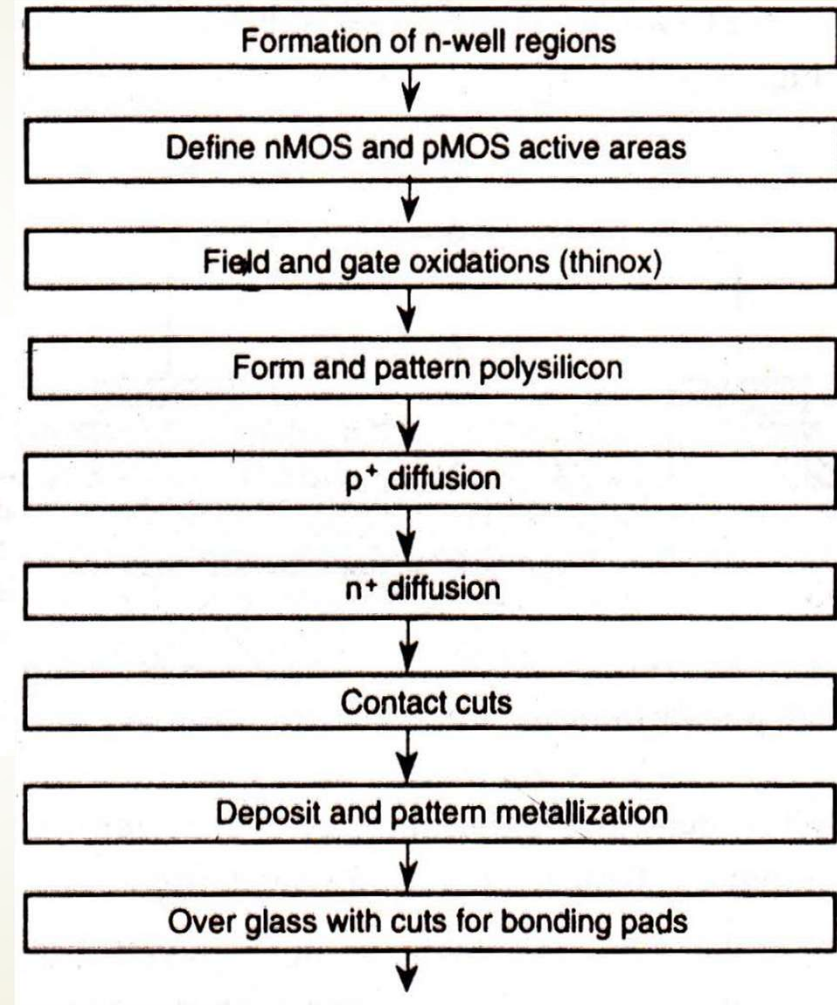
- *Mask 5* - This is usually performed using the negative form of the p-plus mask and defines those areas where n-type diffusion is to take place.
- *Mask 6* - Contact cuts are now defined.
- *Mask 7* - The metal layer pattern is defined by this mask.
- *Mask 8* - An overall passivation (overglass) layer is now applied and Mask 8 is needed to define the openings for access to bonding pads.



N-Well Process

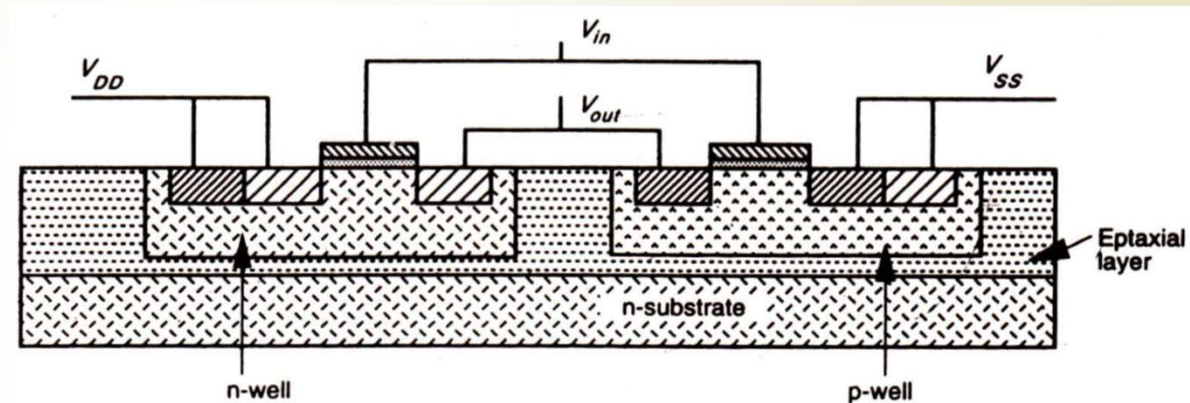


Rohit Lorenzo, VIT-AP

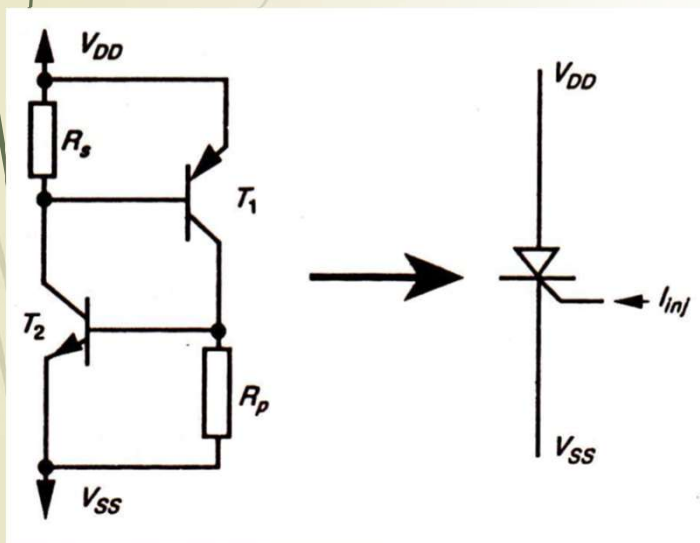


Twin-Tub Process

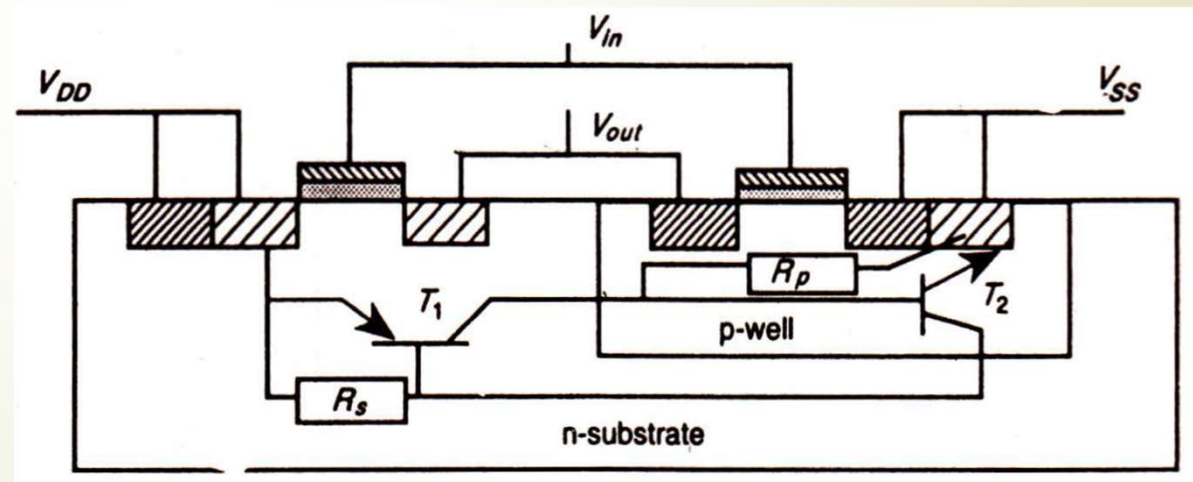
- ▶ A logical extension of the p-well and n-well approaches is the twin-tub fabrication process. Here we start with a substrate of high resistivity n-type material and then create both n-well and p-well regions.
- ▶ Through this process it is possible to preserve the performance of n-transistors without compromising the p-transistors.
- ▶ Doping control is more readily achieved and some relaxation in manufacturing tolerances results. This is particularly important as far as latch-up is concerned.



Latch-up in CMOS Process



- A problem which is inherent in the p-well and n-well processes is due to the relatively large number of junctions which are formed in these structures and as mentioned earlier, the consequent presence of parasitic transistors and diodes.
- Latch-up is a condition in which . The parasitic components give rise to the establishment of low-resistance conducting paths between V_{DD} and V_{SS} with disastrous results. Careful control during fabrication is necessary to avoid this problem.



Remedies for the latch-up problem include

- ▶ 1. an increase in substrate doping levels with a consequent drop in the value of R_S
- ▶ 2. reducing R_P by control of fabrication parameters and by ensuring a low contact resistance to V_{SS} .
- ▶ 3. other more elaborate measures such as the introduction of guard rings.

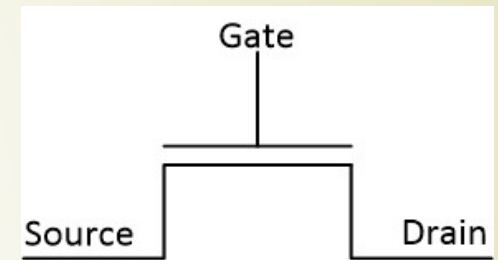
MOS Transistor

- ▶ The Metal oxide semiconductor Field Effect Transistor (MOSFET) is the fundamental building block of CMOS digital integrated circuits.
- ▶ Compared to the bipolar junction transistor (BJT), the MOS transistor occupies a relatively smaller silicon area, and its fabrication involves fewer processing steps.
- ▶ These technological advantages, together with the relative simplicity of MOSFET operation, have helped make the MOS transistor the most widely used switching device in VLSI circuits.
- ▶ The nMOS transistor is used as the primary switching device in virtually all digital circuit applications, whereas the pMOS transistor is used mostly in conjunction with the nMOS device in CMOS circuits. However, the basic operation principles of both nMOS and pMOS transistors are very similar to each other.

Rohit Lorenzo, VIT-AP

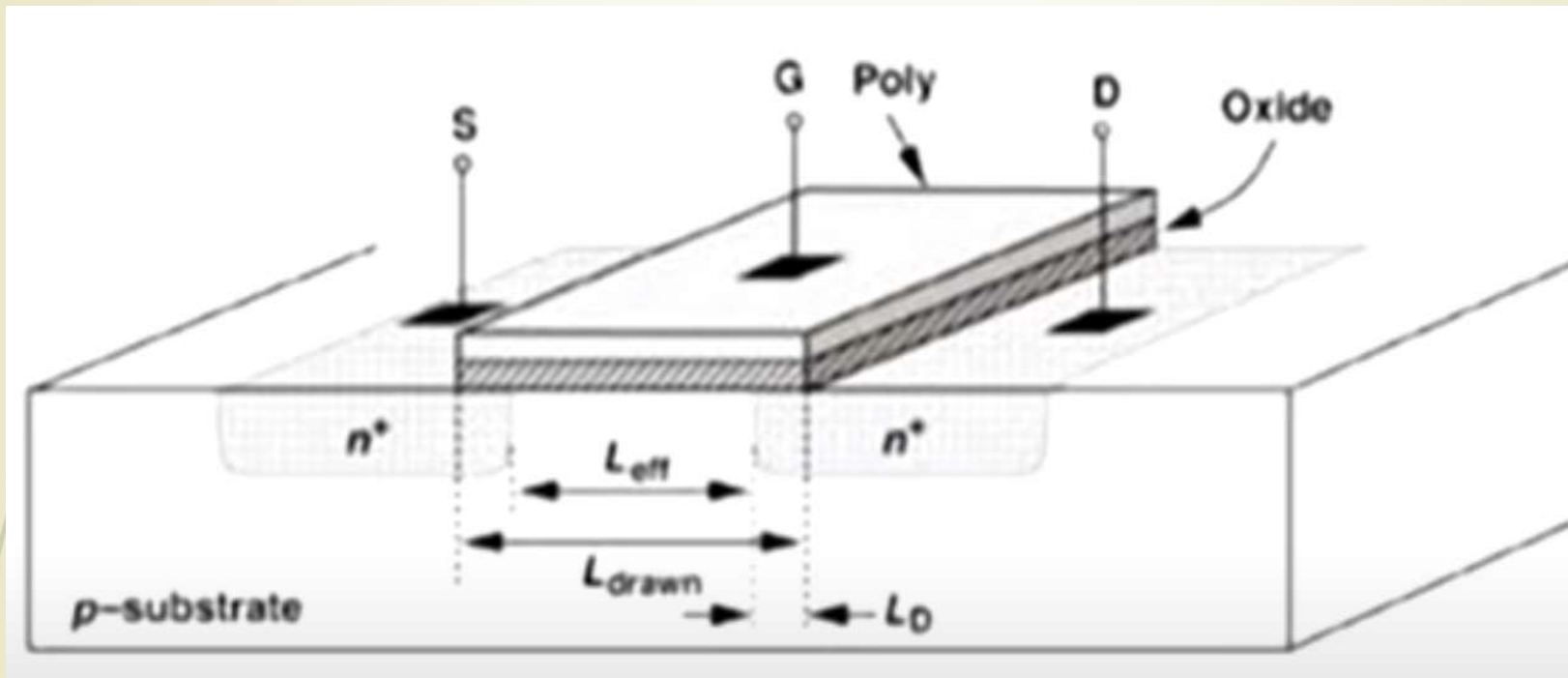
MOSFET as a switch

- MOSFET can be considered as a switch which operates with proper biasing.
- Field Effect Transistor – The flow of electron is guided by amount of electric field present in the device.
- Biasing – The amount of voltage applied to source , gate and drain.
- This helps to give many answers itself (Questions)
 - 1. For what value of gate voltage device will turn ON (Threshold volatge)
 - 2. What is the resistance between source and drain when device is turn ON (OFF) .
 - 3. What limit the speed of the device.



Rohit Lorenzo, VIT-AP

MOSFET Structure

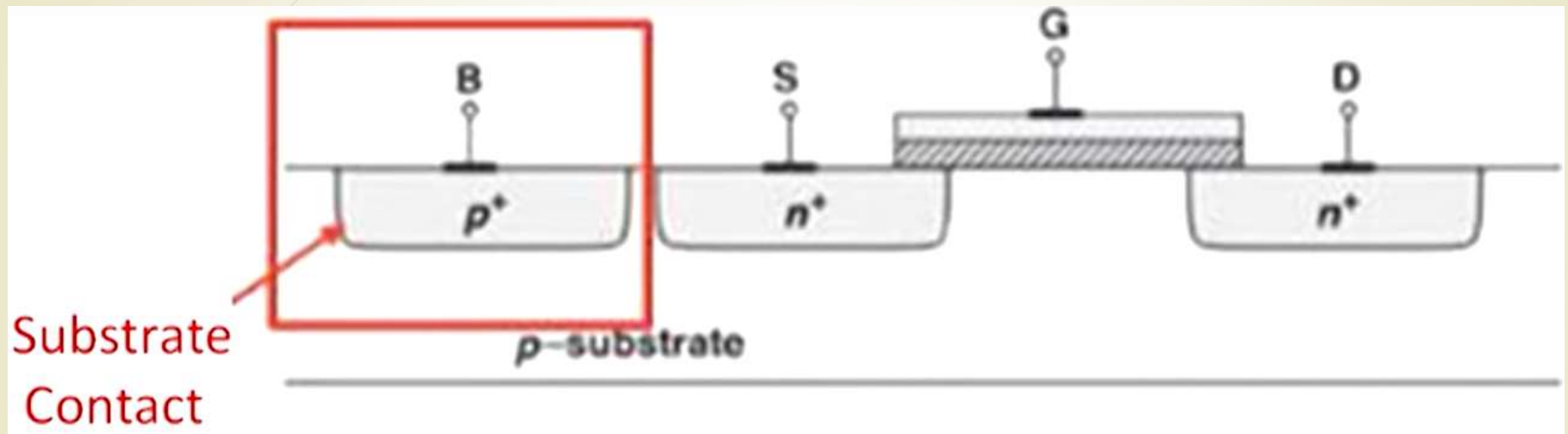


Rohit Lorenzo, VIT-AP

L_D : Side Diffusion Length

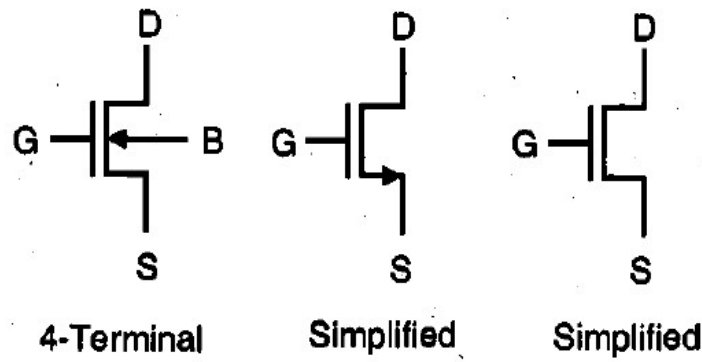
$$L_{eff} = L_{drawn} - 2L_D$$

Body terminal

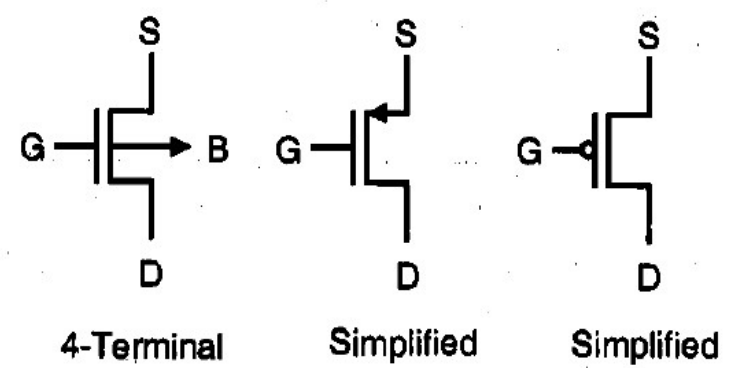


- The substrate bias should be connected with the negative most supply in case of NMOS and Positive most supply in case of PMOS.

MOS symbol



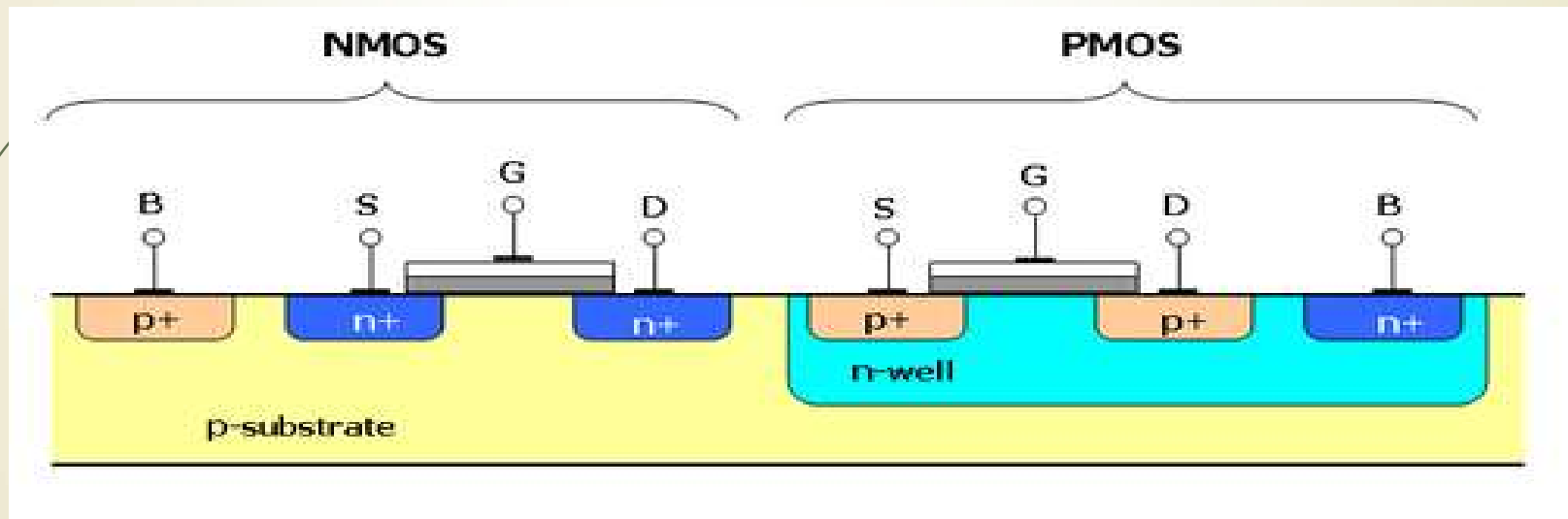
n-channel MOSFET



p-channel MOSFET

n-Well Process

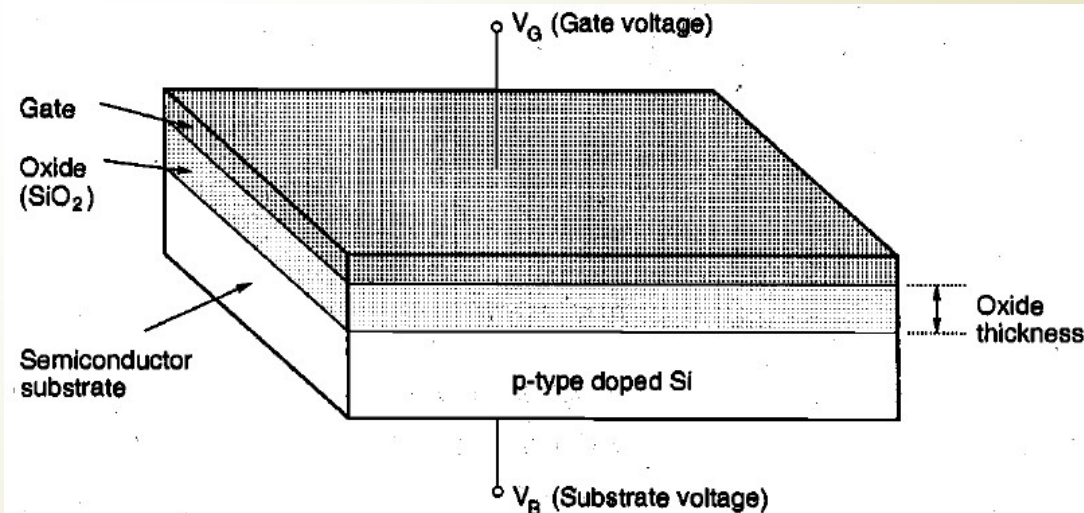
- NMOS and PMOS are in general made in same wafer, in which one device can be placed in local substrate called as well



Rohit Lorenzo, VIT-AP

MOS Structure or MOS capacitor

- ▶ The MOS structure forms a capacitor, with the gate and the substrate acting as the two terminals (plates) and the oxide layer as the dielectric.
- ▶ The carrier concentration and its local distribution within the semiconductor substrate can now be manipulated by the external voltages applied to the gate and substrate terminals.
- ▶ A basic understanding of the bias conditions for establishing different carrier concentrations in the substrate will also provide valuable insight into the operating conditions



Continued....

- ▶ The equilibrium concentrations of mobile carriers in a semiconductor always obey the Mass Action Law given by

$$n \cdot p = n_i^2 \quad \text{..... (1)}$$

- ▶ Here, **n** and **p** denote the mobile carrier concentrations of electrons and holes, respectively, and **n_i** denotes the intrinsic carrier concentration of silicon. Assuming that the substrate is uniformly doped with an acceptor (e.g., Boron) concentration **N_A**, the equilibrium electron and hole concentrations in the p-type substrate are approximated by

$$p_{po} \cong n_A$$

$$n_{po} \cong \frac{n_i^2}{n_A} \quad \text{..... (2)}$$

Energy Band Diagram

- ▶ The location of the equilibrium Fermi level E_F within the band-gap is determined by the doping type and the doping concentration in the silicon substrate.
- ▶ The Fermi potential ϕ_F , which is a function of temperature and doping, denotes the difference between the intrinsic Fermi level E_i , and the Fermi level E_F

$$\phi_F \cong \frac{E_F - E_i}{q} \quad \dots\dots (3)$$

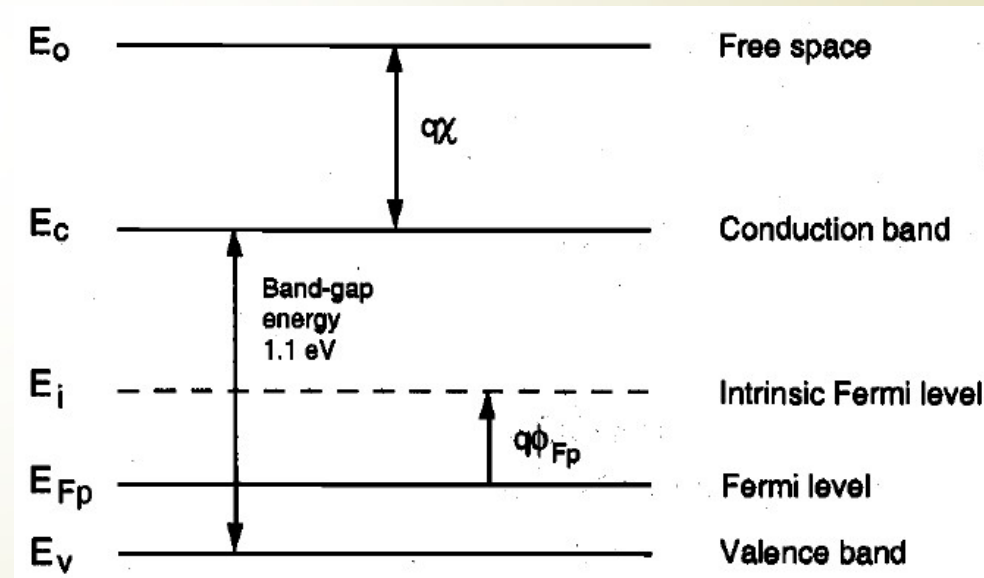
- ▶ For a p-type semiconductor, the Fermi potential can be approximated by

$$\phi_{Fp} = \frac{KT}{q} \ln \frac{n_i}{n_A} \quad \dots\dots (4)$$

- ▶ whereas for an n-type semiconductor (doped with a donor concentration N_D), the Fermi potential is given by

$$\phi_{Fn} = \frac{KT}{q} \ln \frac{n_D}{n_i} \quad \dots\dots (5)$$

Rohit Lorenzo, VIT-AP



Energy Band diagram of P Type silicon substrate

Continued....

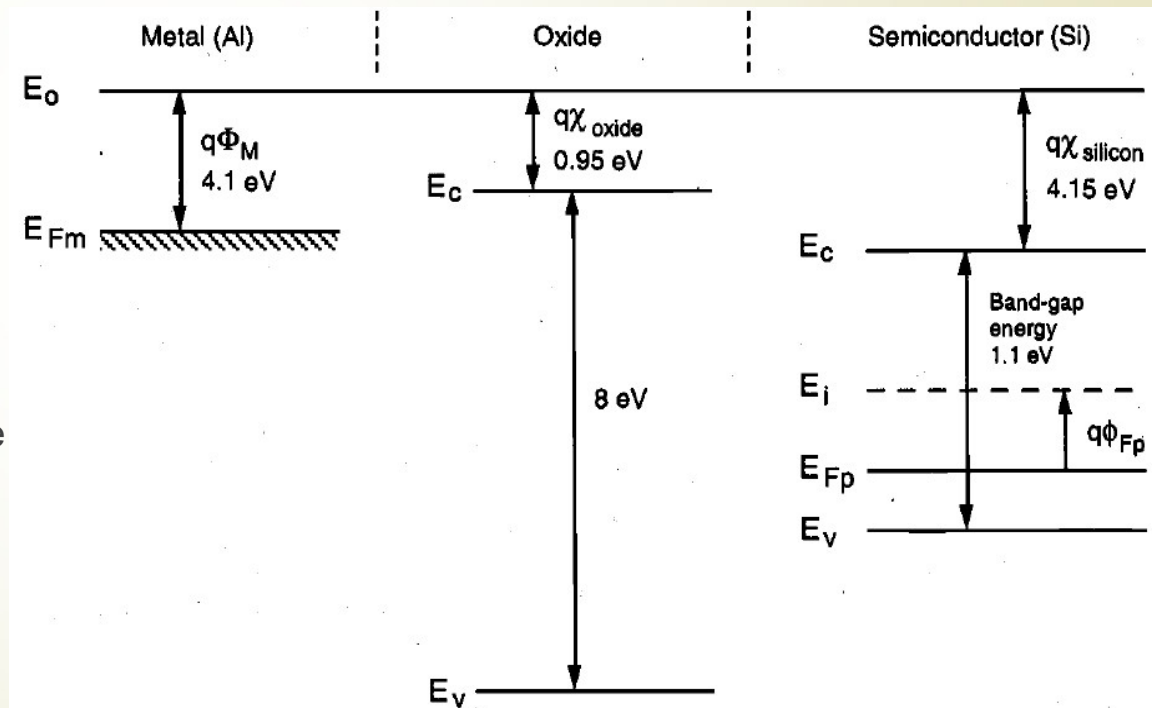
- ▶ The electron affinity of silicon, which is the potential difference between the conduction band level and the vacuum (free-space) level, is denoted by $q\chi$.
- ▶ The energy required for an electron to move from the Fermi level into free space is called the work function $q\phi_s$, and is given by

$$q\phi_s = q\chi + (E_c - E_F) \quad \text{.....(6)}$$

Energy Band Diagram of MOS

- Now consider that the three components of the ideal MOS system are brought into physical contact. The Fermi levels of all three materials must line up, as they form the MOS capacitor
- Because of the work-function difference between the metal and the semiconductor, a voltage drop occurs across the MOS system.
- Part of this built-in voltage drop occurs across the insulating oxide layer.
- The rest of the voltage drop (potential difference) occurs at the silicon surface next to the silicon-oxide interface, forcing the energy bands of silicon to bend in this region

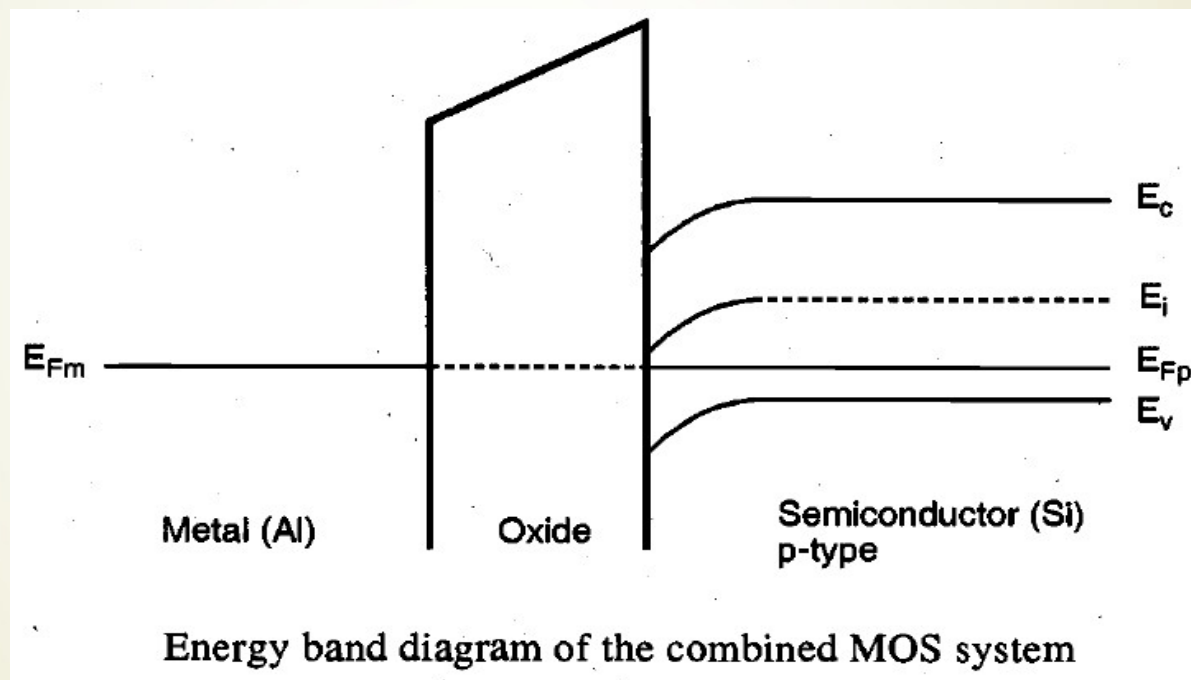
Rohit Lorenzo, VIT-AP



Energy band diagrams of the components that make up the MOS system.

Continued....

- ▶ The resulting combined energy band diagram of the MOS system as shown in figure. Notice that the equilibrium Fermi levels of the semiconductor (Si) substrate and the metal gate are at the same potential.



Problem 1

- Consider the MOS structure that consists of a p-type doped silicon substrate, a silicon dioxide layer, and a metal (aluminum) gate. The equilibrium Fermi potential of the doped silicon substrate is given as $q\phi_{fp} = 0.2eV$. calculate the built-in potential difference across the MOS system. Assume that the MOS system contains no other charges in the oxide or on the silicon-oxide interface?

Solution

- we have to calculate the work function for the doped silicon, the electron affinity of silicon is **4.15 eV**, the work function $q\phi_s$ is found

$$q\phi_s = q\chi + (E_c - E_F) = 4.15eV + 0.75eV = 4.9eV$$

- calculate the work function difference between the silicon substrate and the aluminum gate. Note that the work function of aluminum is given as 4.1 eV. the built-in potential difference across this MOS system

$$q\phi_M - q\phi_s = 4.1eV - 4.9eV = -0.8eV$$

- If a voltage corresponding to this potential difference is applied externally between the gate and the substrate, the bending of the energy bands near the surface can be compensated, i.e., the energy bands become "flat."

Thus, the voltage defined by

$$V_{FB} = \Phi_M - \Phi_S$$

is called **flat band voltage**

The MOS System under External Bias

- ▶ We now turn our attention to the electrical behavior of the MOS structure under externally applied bias voltages. Depending on the polarity and the magnitude of V_G , three different operating regions can be observed for the MOS system: accumulation, depletion, and inversion.
- ▶ The majority carrier concentration near the surface becomes larger than the equilibrium hole concentration in the substrate; hence, this condition is called *carrier accumulation on the surface*.
- ▶ Note that in this case, the oxide electric field is directed towards the gate electrode.
- ▶ The negative surface potential also causes the energy bands to bend upward near the surface.

Rohit Lorenzo, VIT-AP

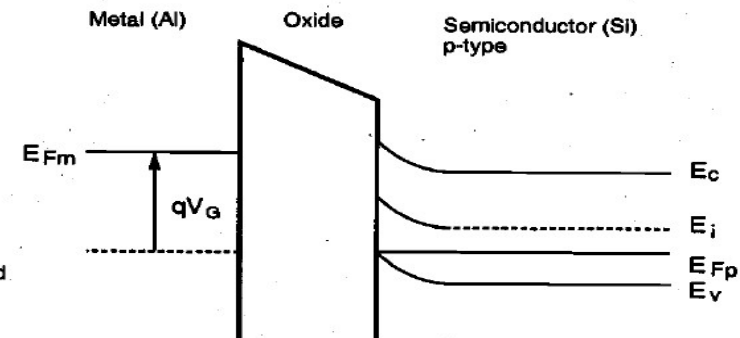
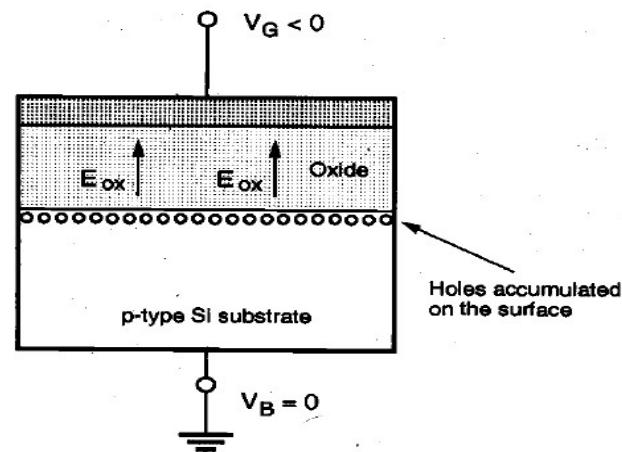
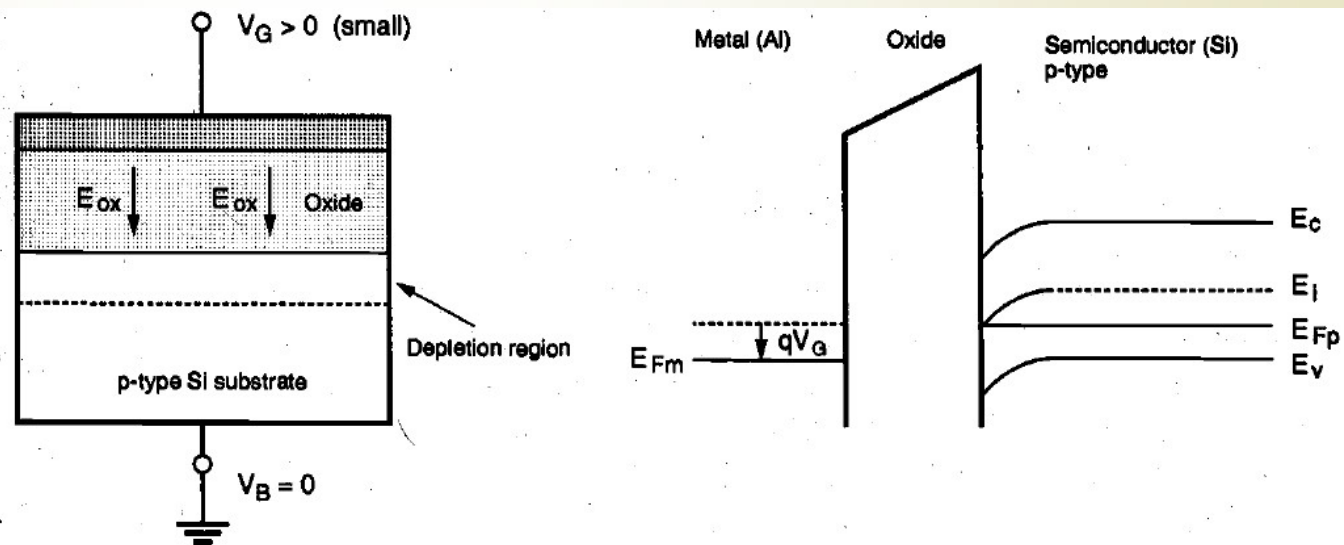


Figure The cross-sectional view and the energy band diagram of the MOS structure operating in accumulation region.

Depletion Mode under Small External Bias

- Now a small positive gate bias V_G is applied to the gate electrode. Since the substrate bias is zero, the oxide electric field will be directed towards the substrate in this case. The positive surface potential causes the energy bands to bend downward near the surface.
- The majority carriers, i.e., the holes in the substrate, will be repelled back into the substrate as a result of the positive gate bias, and these holes will leave negatively charged fixed acceptor ions behind.
- Thus, a depletion region is created near the surface.



Rohit Lorenzo, VIT-AP

Figure The cross-sectional view and the energy band diagram of the MOS structure operating in depletion mode, under small gate bias.

Calculation of depletion thickness

- ▶ The thickness x_d of this depletion region on the surface can easily be found as a function of the surface potential ϕ_s .
- ▶ Assume that the mobile charge in a thin horizontal layer parallel to the surface is

$$dQ = -q \cdot N_A \cdot dx \quad \text{..... (7)}$$

- ▶ The change in surface potential required to displace this charge sheet dQ by a distance x_d away from the surface can be found by using the Poisson equation

$$d\phi_s = -x \cdot \frac{dq}{\epsilon_{si}} = \frac{q \cdot N_A \cdot x}{\epsilon_{si}} dx \quad \text{.....(8)}$$

- ▶ Integrating along the vertical direction

$$\int_{\phi_F}^{\phi_s} d\phi_s = \int_0^{x_d} \frac{q \cdot N_A \cdot x}{\epsilon_{si}} dx \quad \text{..... (9)}$$

Continued...



$$\varphi_S - \varphi_F = \frac{q \cdot N_A \cdot x_d^2}{2\epsilon_{si}} \quad \text{.....(10)}$$

- Thus, the depth of the depletion region is

$$x_d = \sqrt{\frac{2\epsilon_{si} \cdot |\varphi_S - \varphi_F|}{q \cdot N_A}} \quad \text{.....(11)}$$

- the depletion region charge density, which consists solely of fixed acceptor ions in this region, is given by the following expression

$$Q = -q \cdot N_A \cdot x_d = \sqrt{2q \cdot N_A \epsilon_{si} \cdot |\varphi_S - \varphi_F|} \quad \text{..... (12)}$$

Inversion Mode under increase External Bias

- consider next a further increase in the positive gate bias. As a result of the increasing surface potential, the downward bending of the energy bands will increase as well.
- Eventually, the mid-gap energy level E_i becomes smaller than the Fermi level E_{Fp} on the surface, which means that the substrate semiconductor in this region becomes n-type.
- The positive gate potential attracts additional minority carriers (electrons) from the bulk substrate to the surface.
- The n-type region created near the surface by the positive gate bias is called the *inversion layer*

Rohit Lorenzo, VIT-AP

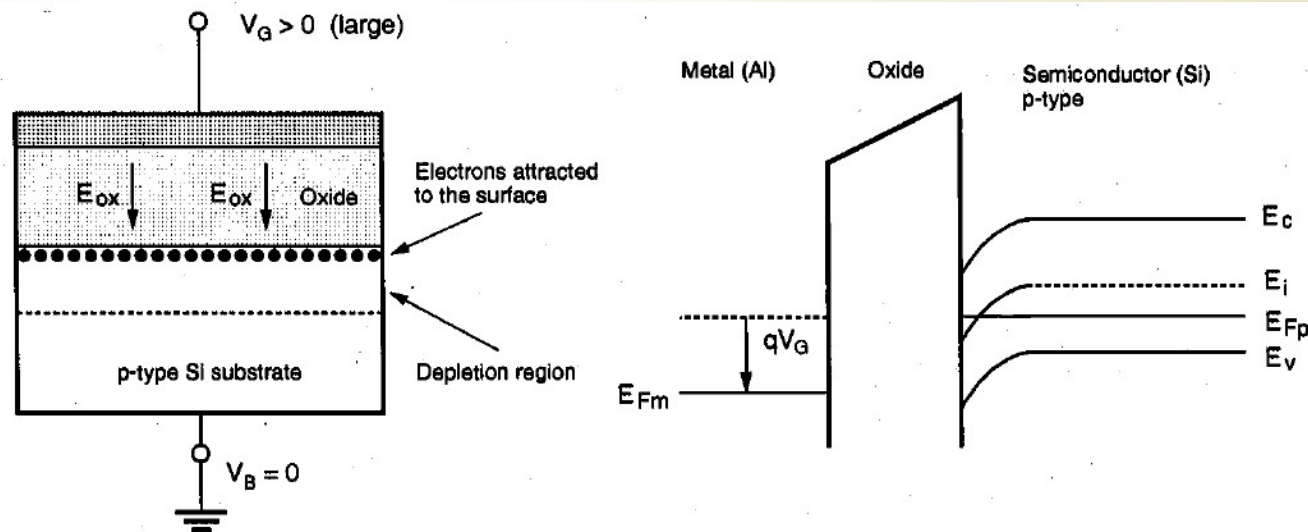


Figure The cross-sectional view and the energy band diagram of the MOS structure in surface inversion, under larger gate bias voltage.

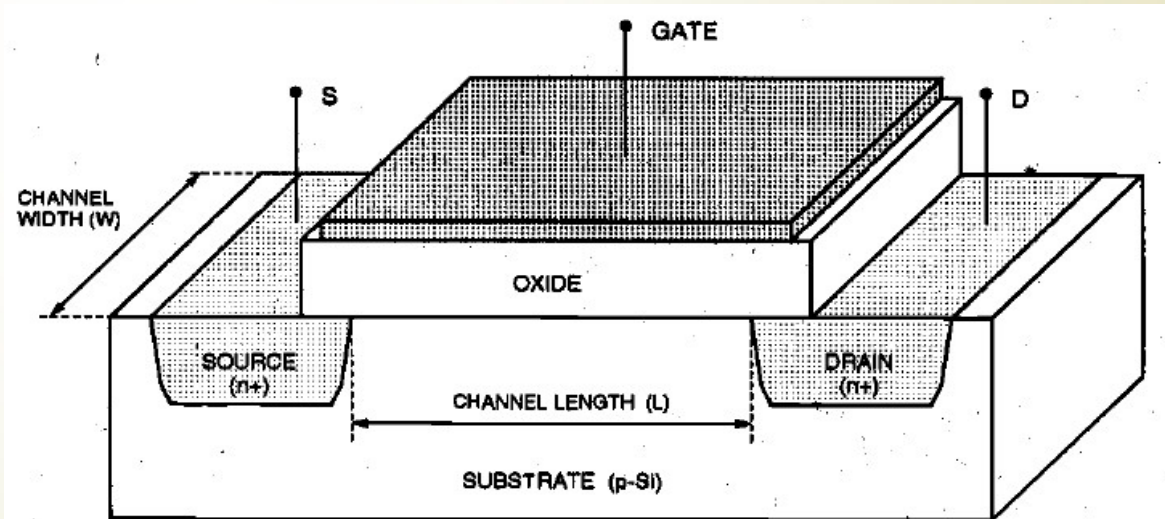
Continued....

- ▶ The surface with a large mobile electron concentration can be utilized for conducting current between two terminals of the MOS transistor.
- ▶ As a practical definition, the surface is said to be inverted when the density of mobile electrons on the surface becomes equal to the density of holes in the bulk (p-type) substrate.
- ▶ Thus, the depletion region depth achieved at the onset of surface inversion is also equal to the maximum depletion depth, x_{dm} , which remains constant for higher gate voltages.
- ▶ Using the inversion condition $\phi_s = -\phi_F$
- ▶ The maximum depletion region depth

$$x_{dm} = \sqrt{\frac{2\epsilon_{si}|\phi_F|}{q \cdot N_A}} \quad \text{.....(13)}$$

MOSFET operation

- ▶ The simple operation principle of this device is: control the current conduction between the source and the drain, using the electric field generated by the gate voltage as a control variable.
- ▶ Since the current flow in the channel is also controlled by the drain-to-source voltage and by the substrate voltage, the current can be considered a function of these external terminal voltages.
- ▶ We will examine in detail the functional relationships between the channel current (also called the *drain current*) and the terminal voltages

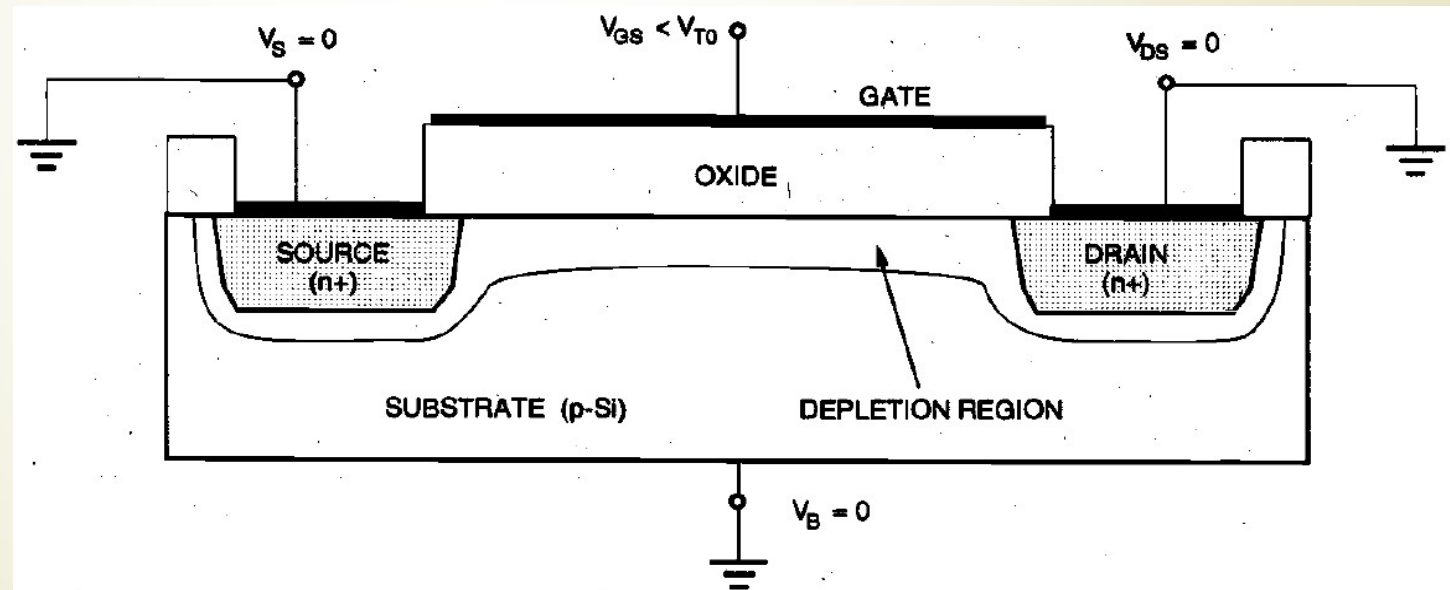


Rohit Lorenzo, VIT-AP

The physical structure of an n-channel enhancement-type MOSFET.

Small V_{GS}

- ▶ The simplest bias condition that can be applied to the n-channel enhancement-type MOSFET.
- ▶ The source, the drain, and the substrate terminals are all connected to ground. A positive gate-to-source voltage V_{GS} is then applied to the gate in order to create the conducting channel underneath the gate.
- ▶ For small gate voltage levels, the majority carriers (holes) are repelled back into the substrate, and the surface of the p-type substrate is depleted.



Rohit Lorenzo, VIT-AP

Figure Formation of a depletion region in an n-channel enhancement-type MOSFET

Increased V_{GS}

- Now assume that the V_{GS} is further increased. As soon as the surface potential in the channel region reaches $-2\phi_F$ surface inversion will be established, and a conducting n-type layer will form between the source and the drain diffusion regions.
- The value of the gate-to-source voltage V_{GS} needed to cause surface inversion to create the conducting channel) is called the threshold voltage V_{T0} .
- $V_{GS} < V_{T0}$ is not sufficient to establish an inversion layer thus, the MOSFET can conduct no current between its source and drain terminals unless $V_{GS} > V_{T0}$.

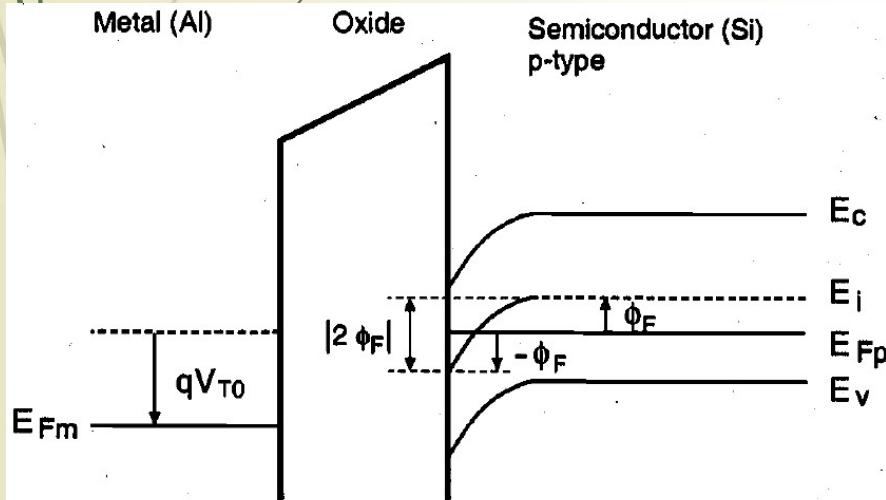


Figure Band diagram of the MOS structure

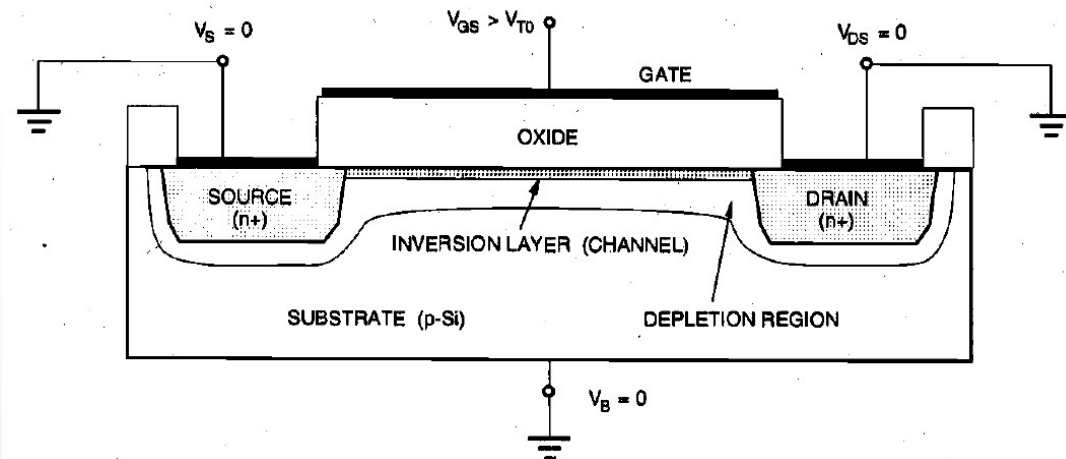


Figure Formation of an inversion layer (channel)

Threshold Voltage

- ▶ The threshold voltage is defined as the gate voltage at which the channel is inverted or form.
- ▶ Physical parameters affecting the threshold voltage V_{T0} are –
 - ▶ (i) the work function difference between the gate and the channel
 - ▶ (ii) the gate voltage component to change the surface potential
 - ▶ (iii) the gate voltage component to offset the depletion region charge
 - ▶ (IV) the voltage component to offset the fixed charges in the gate oxide and in the silicon-oxide interface.
- ▶ The work function difference Φ_{GC} between the gate and the channel reflects the built-in potential (first component) of the MOS system. Depending on the gate material, the work function difference is –
 - ▶
$$\Phi_{GC} = \Phi_F(Substrate) - \Phi_M \quad \text{for metal gate} \quad \dots\dots(14)$$
- ▶ In second component, the externally applied gate voltage must be changed to achieve surface inversion, i.e., to change the surface potential by $-2\Phi_F$.

Continued....

- ▶ The third includes the gate voltage is necessary to offset the depletion region charge. We can calculate the depletion region charge density at surface inversion ($\Phi_S = -\Phi_F$) using equation (12)

$$Q_{B0} = -\sqrt{2q \cdot N_A \cdot \epsilon_{si} \cdot | - 2\phi_F |} \quad \dots\dots\dots (15)$$

- ▶ if the substrate (body) is biased at a different voltage level than the source. then the depletion region charge density can be expressed as a function of the source-to-substrate voltage V_{SB} .

$$Q_B = -\sqrt{2q \cdot N_A \cdot \epsilon_{si} \cdot | - 2\phi_F + V_{SB} |} \quad \dots\dots\dots (16)$$

- ▶ The component that offsets the depletion region charge is then equal to Φ_B / C_{ox} where C_{ox} is the gate oxide capacitance per unit area.

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \quad \dots\dots\dots (17)$$

Continued....

- ▶ The gate voltage component that is necessary to offset this positive charge at the interface is Q_{ox}/C_{ox} .
- ▶ Now, we can combine all of these voltage components to find the threshold voltage. For zero substrate bias, the threshold voltage V_{T0} is expressed as follows:

$$V_{T0} = \Phi_{GC} - 2\Phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} \quad \dots\dots\dots (18)$$

- ▶ For nonzero substrate bias, threshold voltage can be expressed as

$$V_T = \Phi_{GC} - 2\Phi_F - \frac{Q_B}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} \quad \dots\dots\dots (19)$$

- ▶ The generalized form of the threshold voltage can also be written as

$$V_T = \Phi_{GC} - 2\Phi_F - \frac{Q_{Bo}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} - \frac{Q_B}{C_{ox}} + \frac{Q_{Bo}}{C_{ox}} = \Phi_{GC} - 2\Phi_F - \frac{Q_{Bo}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} - \frac{Q_B - Q_{Bo}}{C_{ox}}$$

$$V_T = V_{T0} - \frac{Q_B - Q_{Bo}}{C_{ox}} \quad \dots\dots(20)$$

Continued....

- From eq. (15) and (16) -

- $$\frac{Q_B - Q_{B0}}{C_{ox}} = -\frac{\sqrt{2q \cdot N_A \cdot \epsilon_{Si}}}{C_{ox}} \cdot (\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}) \quad \dots\dots\dots (21)$$

- $$V_T = V_{T0} + \gamma(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}) \quad \dots\dots\dots (22)$$

Where the parameter

$$\gamma = \frac{\sqrt{2q \cdot N_A \cdot \epsilon_{Si}}}{C_{ox}}$$

where γ is the substrate-bias (or body-effect) coefficient

Continued.....

- ▶ The substrate Fermi potential Φ_F is negative in NMOS, positive in PMOS.
- ▶ The depletion region charge densities Q_{BO} and Q_B are negative in PMOS, positive in PMOS.
- ▶ The substrate bias coefficient γ is positive in NMOS, negative in PMOS.
- ▶ The substrate bias voltage V_{SB} is positive in NMOS, negative in PMOS.
- ▶ Typically, the threshold voltage of an enhancement-type n-channel MOSFET is a **positive** quantity, whereas the threshold voltage of a p-channel MOSFET is **negative**.
- ▶ The threshold voltage can be adjusted by selective dopant ion implantation into the channel region of the MOSFET.
- ▶ For nchannel MOSFETs, the threshold voltage is increased (made more positive) by adding extra p-type impurities (acceptor ions). Alternatively, the threshold voltage of the nchannel MOSFET can be decreased (made more negative) by implanting n-type impurities (dopant ions) into the channel region

Problem 2

- Calculate the threshold voltage V_{TO} at $V_{SB} = 0$, for a polysilicon gate n-channel MOS transistor, with the following parameters: substrate doping density $N_A = 10^{16} \text{ cm}^{-3}$, polysilicon gate doping density $N_D = 2 \times 10^{20} \text{ cm}^{-3}$, gate oxide thickness $t_{ox} = 500 \text{ \AA}$, fermi potential of gate $\Phi_F = 0.55V$ and oxide-interface fixed charge density $N_{ox} = 4 \times 10^{10} \text{ cm}^{-2}$.

- Solution**

- First, calculate the Fermi potentials for the p-type substrate and for the n-type polysilicon gate

$$\Phi_{F(\text{substrate})} = \frac{KT}{q} \ln \frac{n_i}{n_A} = 0.026V \cdot \ln \left(\frac{1.45 \cdot 10^{10}}{10^{16}} \right) = -0.35V$$

- work function difference between the gate and the channel

$$\Phi_{GC} = \Phi_F(\text{Substrate}) - \Phi_F(\text{gate}) = -0.35V - 0.55V = -0.90V$$

- The depletion region charge density at $V_{SB} = 0$ is found as follows:

$$Q_{B0} = -\sqrt{2q \cdot N_A \cdot \epsilon_{si} \cdot | - 2\phi_F(\text{Substrate}) |}$$

Continued.....

$$Q_{B0} = -\sqrt{2 \times 1.6 \times 10^{-19} \times 10^{16} \times 11.7 \times 8.85 \times 10^{-14}} - 2 \times 0.35| \\ = 4.82 \times 10^{-8} \text{C/cm}^2$$

- ▶ The oxide-interface charge is:

$$Q_{ox} = q \cdot N_{ox} = 1.6 \times 10^{-19} \times 4 \times 10^{10} \text{cm}^{-2} \\ = 6 \times 4 \times 10^{-9} \text{C/cm}^2$$

- ▶ The gate oxide capacitance per unit area is calculated using the dielectric constant of silicon dioxide and the oxide thickness t_{ox} .

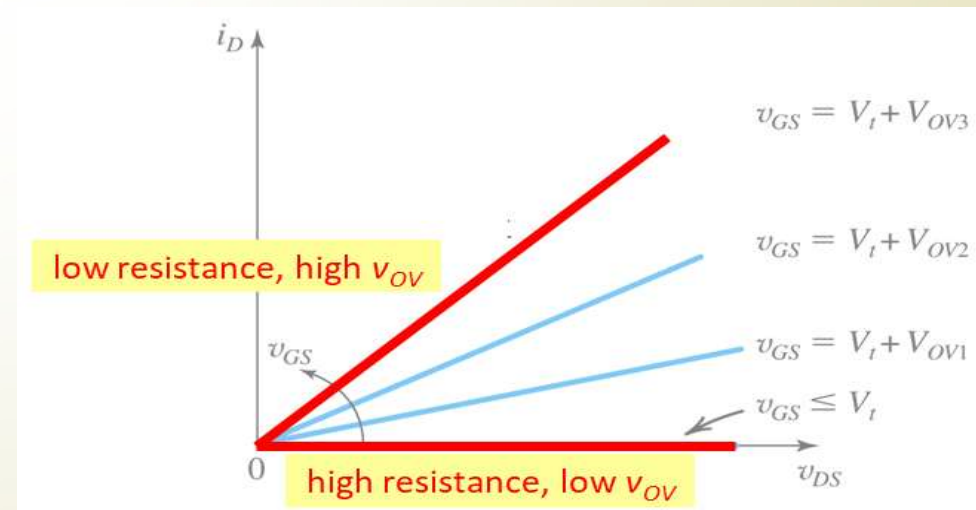
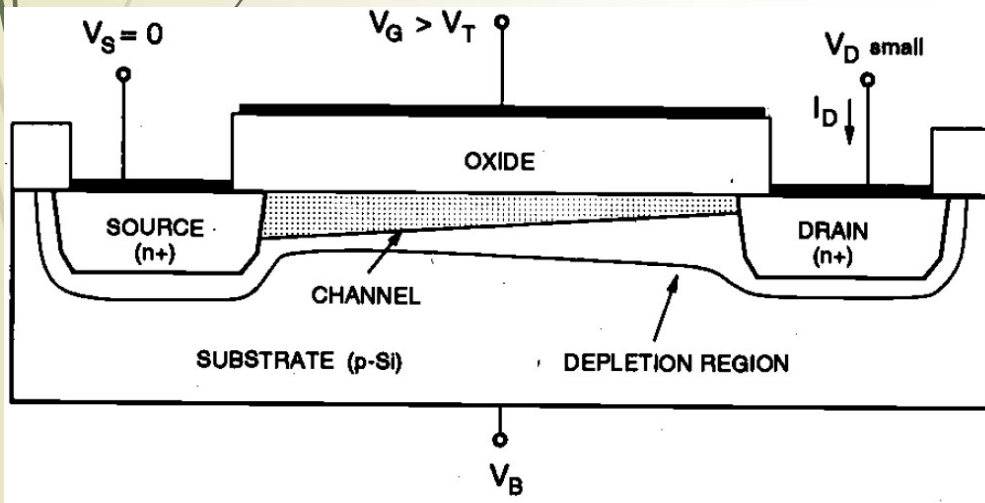
$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.97 \times 8.85 \times 10^{-14} \text{F/cm}}{500 \times 10^{-8} \text{cm}} = 7.03 \times 10^{-8} \text{F/cm}^2$$

- ▶ Now, we can combine all components and calculate the threshold voltage.

$$V_{T0} = \Phi_{GC} - 2\Phi_{F(substrate)} - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} = 0.90 - (-0.70) - (-0.69) - 0.09 = 0.40 \text{V}$$

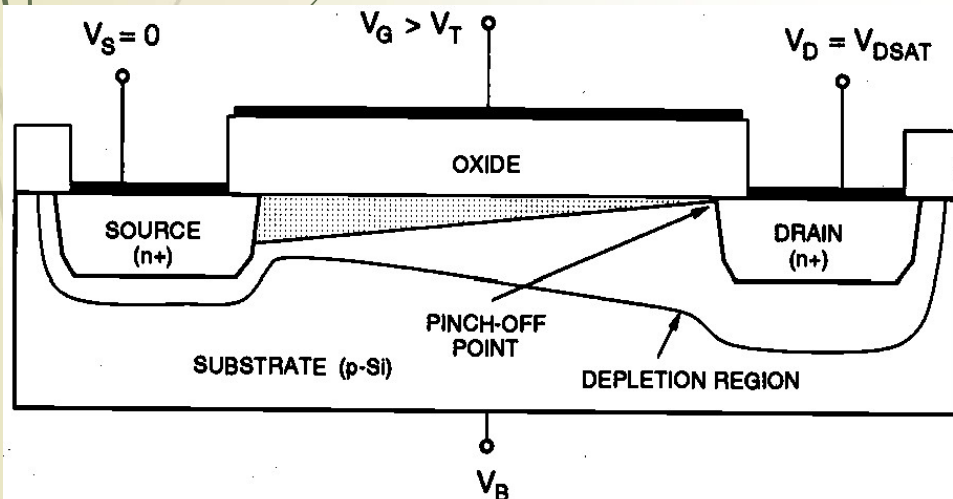
MOSFET Operation: A Qualitative View

- Small V_{DS} (Operating in linear or triode region)
- At $V_{GS} > V_T$ the inversion layer is established on the surface, an n-type conducting channel forms between the source and the drain, which is capable of carrying the drain current.
- If a small drain voltage $V_{DS} > 0$ is applied, a drain current proportional to V_{DS} will flow from the source to the drain through the conducting channel. This operation mode is called the **linear region**. Thus, in linear region operation, the channel region acts as a voltage-controlled resistor. The electron velocity in the channel for this case is usually much lower than the drift velocity limit.

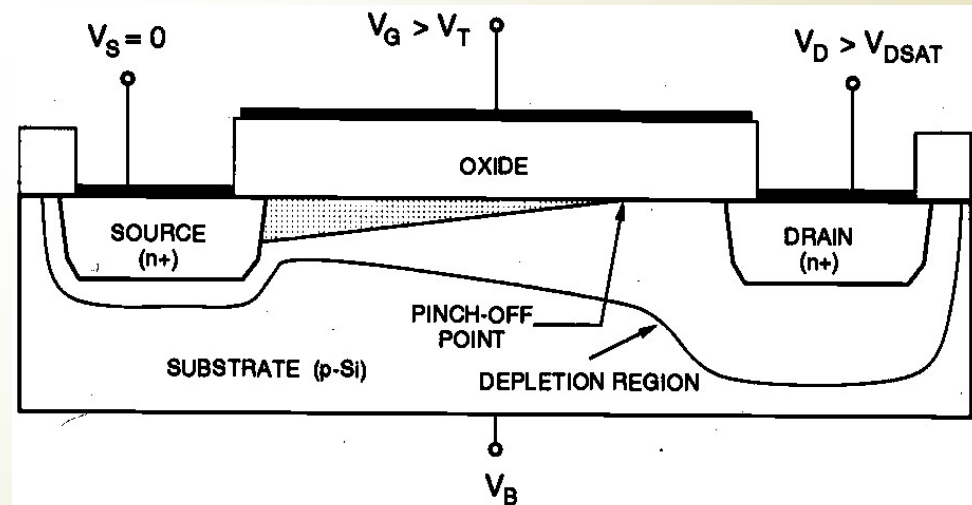


Increased V_{DS}

- As the drain voltage is increased, the inversion layer charge and the channel depth at the drain end start to decrease. Eventually, for $V_{DS} = V_{DSAT}$, the inversion charge at the drain is reduced to zero, which is called the **pinch-off point**.
- Beyond the pinch-off point, $V_{DS} > V_{DSAT}$ a depleted surface region forms adjacent to the drain. This operation mode of the MOSFET is called the saturation mode or the saturation region.



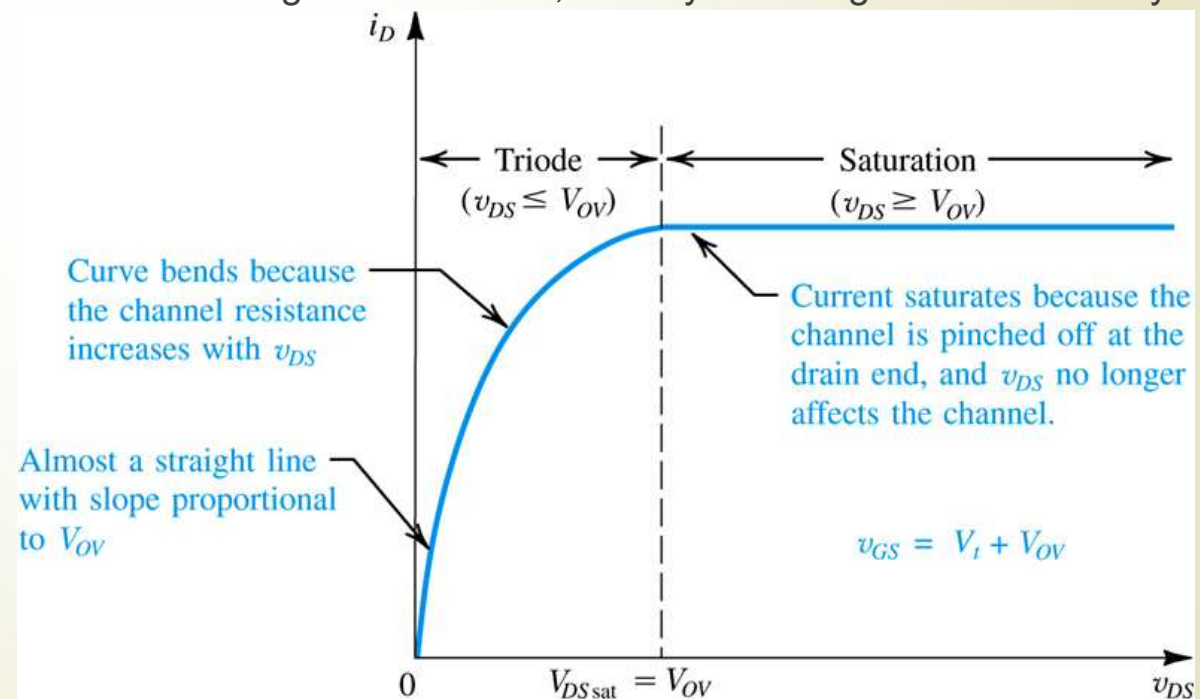
Operating at the edge of Saturation



Operating beyond saturation

Continue.....

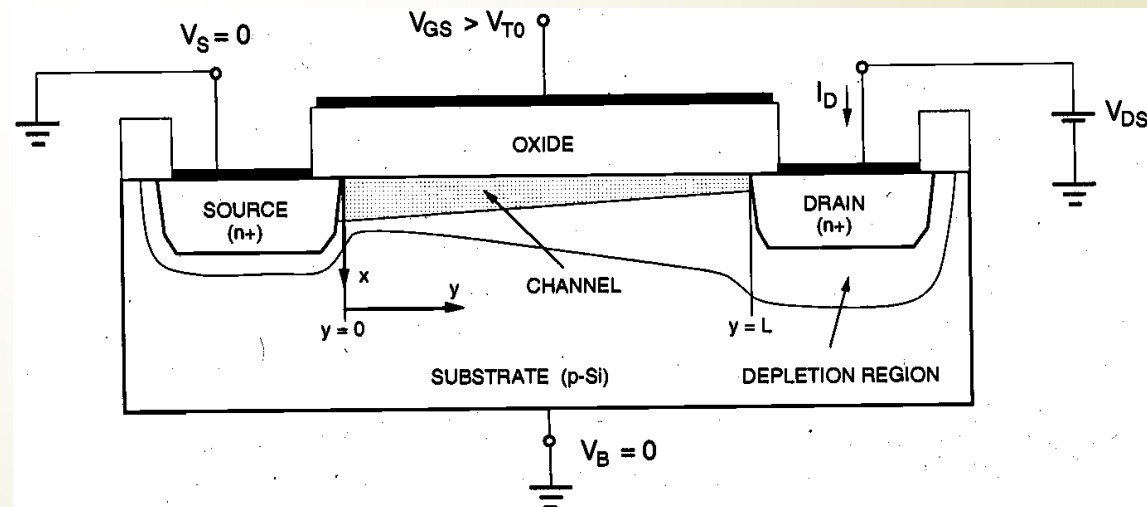
- For a MOSFET operating in the saturation region, the effective channel length is reduced as the inversion layer near the drain vanishes.
- Note that the pinched-off (depleted) section of the channel absorbs most of the excess voltage drop ($V_{DS} - V_{DSAT}$). Electrons arriving from the source to the channel-end are injected into the drain-depletion region and are accelerated toward the drain in this high electric field, usually reaching the drift velocity limit.



MOSFET current voltage characteristics & Equation

- we will use the gradual channel approximation (GCA) for establishing the MOSFET current-voltage relationships.
- Gradual Channel Approximation**
- consider the cross-sectional view of the n-channel MOSFET operating in the linear mode.
- We define the coordinate system for this structure such that the x-direction is perpendicular to the surface, pointing down into the substrate, and the y-direction is parallel to the surface.
- The y-coordinate origin ($y = 0$) is at the source end of the channel. The channel voltage with respect to the source will be denoted by $V_c(y)$.
- Now assume that the threshold voltage .
- is constant along the entire channel region,
- between $y = 0$ and $y = L$.

Rohit Lorenzo, VIT-AP



Cross-sectional view of an n-channel transistor, operating in linear region

Continue.....

- Next, assume that the electric field component $\mathbf{E}_y$ along the y-coordinate is dominant compared to the electric field component $\mathbf{E}_x$ along the x-coordinate.

- Note that the boundary conditions for the channel voltage $\mathbf{V}_C$ are:

$$V_C(y = 0) = V_s = 0 \quad \text{..... (23)}$$

$$V_C(y = L) = V_{DS} \quad \text{.....(24)}$$

- Also, it is assumed that the entire channel region between the source and the drain is inverted

$$V_{GD} \geq V_{T0}$$

$$V_{GD} = V_{GS} - V_{DS} \geq V_{T0} \quad \text{.....(25)}$$

- Let $\mathbf{Q}_I(\mathbf{y})$ be the total mobile electron charge in the surface inversion layer. This charge can be expressed as a function of $\mathbf{V}_{GS}$ and $\mathbf{V}_C(\mathbf{y})$ as follows –

$$Q_I(y) = C_{ox}[V_{GS} - V_C(y) - V_{T0}] \quad \text{...(26)}$$

Continue.....

- Figure shows the spatial geometry of the surface inversion layer and indicates its significant dimensions.
- Now consider the incremental resistance dR . Assume constant surface mobility μ_n . Minus sign is due to negative charge Q_I .

$$dR = -\frac{dy}{W \cdot \mu_n \cdot Q_I(y)} \quad \dots\dots\dots(27)$$

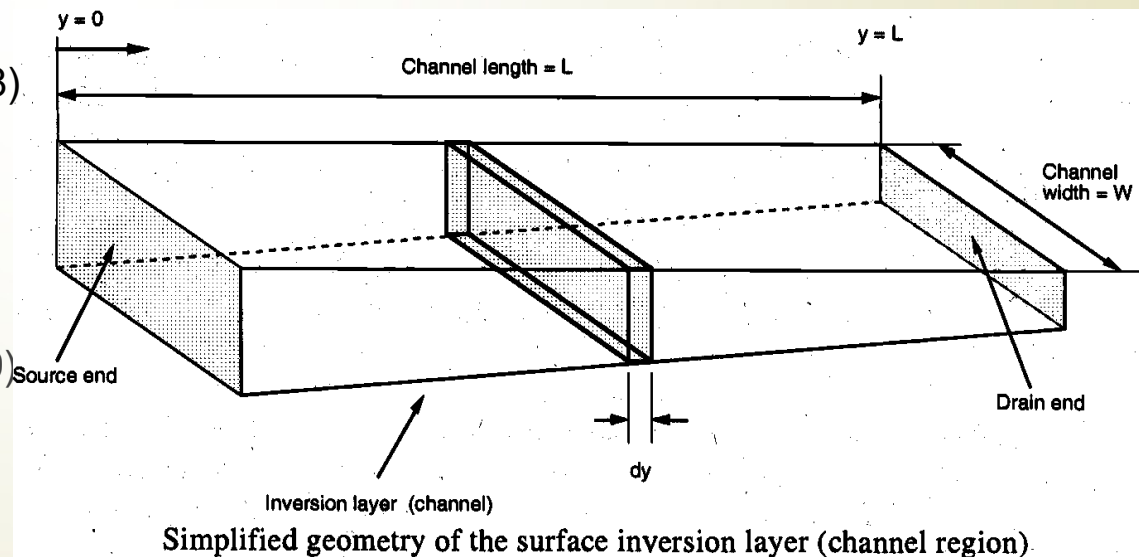
- voltage drop along the incremental segment dy , in the y-direction.

$$dV_C = I_D \cdot dR = -\frac{I_D}{W \cdot \mu_n \cdot Q_I(y)} \cdot dy \quad \dots(28)$$

- This equation can now be integrated along the channel, i.e., from $y = 0$ to $y = L$,

$$\int_0^L I_D \cdot dy = -W \cdot \mu_n \int_0^{V_{Ds}} Q_I(y) \cdot dV_C \quad \dots(29)$$

Rohit Lorenzo, VIT-AP



Continue.....

- Substitute the value of $Q_i(y)$ from eqn (26) to eqn (29).

$$I_D \cdot L = W \cdot \mu_n \cdot C_{ox} \int_0^{V_{DS}} (V_{GS} - V_C - V_{T0}) \cdot dV_C \quad \dots(30)$$

- Assuming that the channel voltage V_C , is the only variable that depends on the position y , the drain current is found as follows.

$$I_D = \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L} [2 \cdot (V_{GS} - V_{T0}) V_{DS} - V_{DS}^2] \quad \dots\dots(31)$$

$$I_D = \frac{K}{2} [2 \cdot (V_{GS} - V_{T0}) V_{DS} - V_{DS}^2] \quad \dots\dots (32)$$

- where the parameters K is defined as

$$K = \frac{\mu_n \cdot C_{ox} W}{L} = K' \frac{W}{L} \quad \text{where} \quad K' = \mu_n \cdot C_{ox}$$

Continue.....

- ▶ Note that, in addition to the process dependent constants **k** and **V**, the current-voltage relationship is also affected by the device dimensions, **W** and **L**.

- ▶ The equation (32) is not valid beyond the linear region/ saturation region boundary.

- ▶
$$V_{DS} \geq V_{DSAT} = V_{GS} - V_T \quad \text{.....(33)}$$

- ▶ The saturation drain current equation can be found simply by substituting Value of V_{DS} to equation (32)

$$I_{D(sat)} = \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L} [2 \cdot (V_{GS} - V_{T0})(V_{GS} - V_{T0}) - (V_{GS} - V_{T0})^2]$$

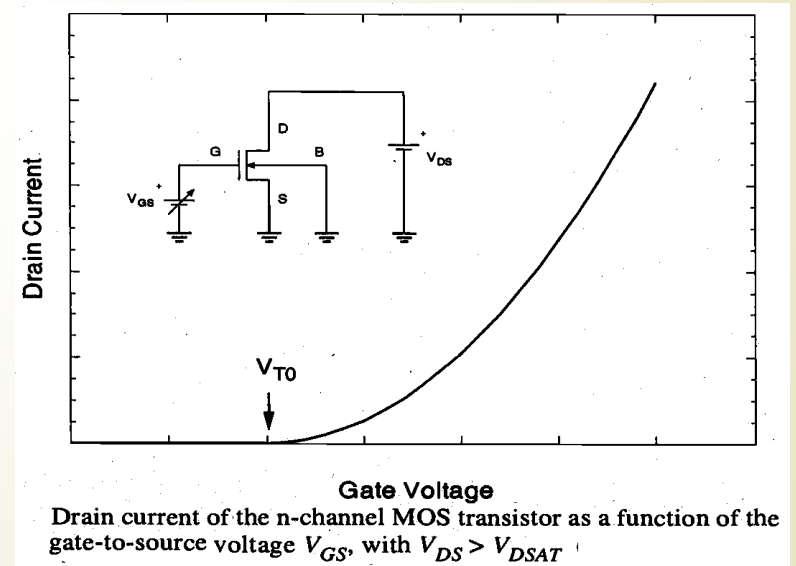
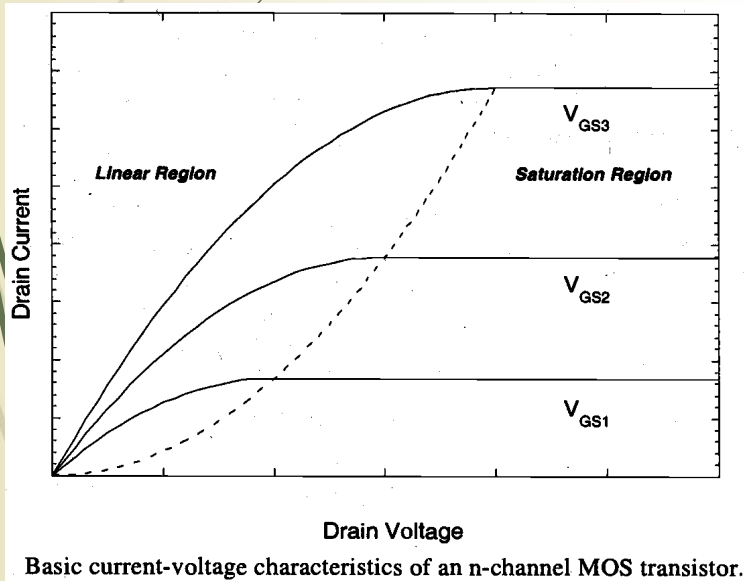
- ▶
$$I_{D(sat)} = \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L} (V_{GS} - V_{T0})^2 \quad \text{.....(34)}$$

$$I_{D(sat)} = \frac{K}{2} (V_{GS} - V_{T0})^2$$

- ▶ The drain current I_D becomes a function only of the gate-to-source voltage V_{GS} , beyond the saturation boundary.

Current-Voltage characteristics

- Figure shows the typical drain current versus drain voltage characteristics of an n-channel MOSFET.
- The parabolic boundary between the linear and the saturation regions is indicated here by the dashed line.
- The current-voltage characteristics of the MOS transistor can also be visualized by plotting the drain current as a function of the gate voltage. The current is obviously equal to zero for any gate voltage smaller than the threshold voltage V_{T0} .



Problem 3

- Consider a process technology for which $L = 0.4 \mu\text{m}$, $t_{ox} = 8 \text{ nm}$, $\mu_n = 450 \text{ cm}^2/\text{V s}$, and $V_t = 0.7 \text{ V}$. Find C_{ox} and For a MOSFET with $W/L = 8 \mu\text{m}/0.8 \mu\text{m}$ calculate the value of V_{GS} and V_{DS} needed to operate the transistor in the saturation region with a dc current $I_D = 100 \mu\text{A}$. For the device in (b), find the value of V_{GS} required to cause the device to operate as a 1000Ω resistor for very small $V_{DS}=1$.

- Solution**

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-11} \text{ F/m}}{8 \times 10^{-9} \text{ m}} = 4.32 \times 10^{-3} \text{ F/m}^2$$

- $K' = \mu_n \cdot C_{ox} = 450(\text{cm}^2/\text{V.s}) \times 4.32 \times 10^{-3}(\text{F/m}^2)$
- $= 450 \times 10^8 (\mu\text{m}^2/\text{V.s}) \times 4.32 \times 10^{-15}(\text{F}/\mu\text{m}^2) = 194 \mu\text{A/V}^2$

- For the operation in saturation mode

$$I_{D(sat)} = K' \cdot \frac{W}{L} (V_{GS} - V_{T0})^2$$

$$100 = \frac{1}{2} \times 194 \times \frac{8}{0.8} (V_{GS} - 0.7)^2$$

Continue.....

➤ $V_{GS} - 0.7 = 0.32V$; $V_{GS} = 1.02V$; $V_{DS} = V_{GS} - V_t = 0.32V$

(c) MOSFET in the triode region with V_{DS} very small

$$I_D = K' \cdot \frac{W}{L} (V_{GS} - V_{T0}) V_{DS}$$

$$r_{DS} = \frac{V_{DS}}{i_{DS}}$$

$$r_{DS} = \frac{1}{K' \cdot \frac{W}{L} (V_{GS} - V_{T0})}$$

$$1000 = \frac{1}{194 \times 10^{-6} \times 10 (V_{GS} - 0.7)}$$

$$V_{GS} - 0.7 = 0.52V$$

$$V_{GS} = 1.22V$$

Rohit Lorenzo, VIT-AP

Channel length modulation

- we will examine the mechanisms of channel pinch-off and current flow in saturation mode in more detail.
- The inversion layer charge at the source end of the channel is

$$Q_I(y = 0) = -C_{ox}[V_{GS} - V_{T0}] \quad \dots\dots\dots(35)$$

- and the inversion layer charge at the drain end of the channel is

$$Q_I(y = L) = -C_{ox}[V_{GS} - V_{T0} - V_{DS}] \quad \dots\dots\dots(36)$$

- Note that at the edge of saturation, i.e., when the drain-to-source voltage reaches V_{DSAT} ,

$$V_{DS} = V_{DSAT} = V_{GS} - V_{T0} \quad \dots\dots\dots(37)$$

- the inversion layer charge at the drain end becomes zero, according to (37). In reality, the channel charge does not become exactly equal to zero because GCA is just a simple approximation, but it indeed becomes very small.

$$Q_I(y = L) \approx 0 \quad \dots\dots\dots(38)$$

- Thus we can state channel is pinched off at the drain end

Channel length modulation

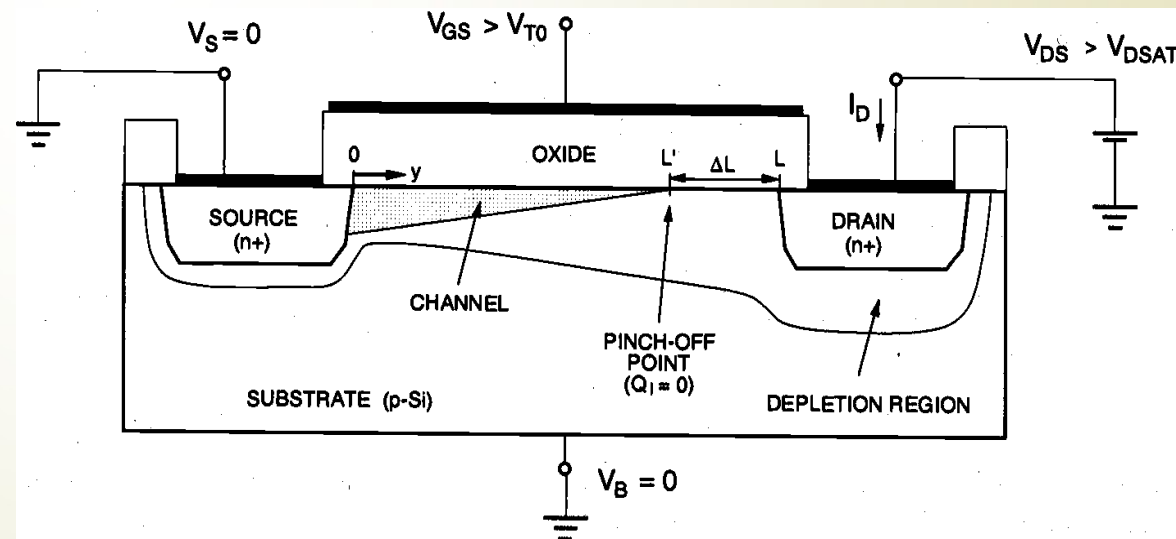
- If the drain-to-source voltage V_{DS} is increased even further beyond the saturation edge so that $V_{DS} > V_{DSAT}$, an *even larger* portion of the channel becomes pinched-off.
- Consequently, the effective channel length is reduced to $L' = L - \Delta L$ (39)
- where ΔL is the length of the channel segment with $Q = 0$. Hence, the pinch off point moves from the drain end of the channel toward the source with increasing V_{DS} . Since $Q(y) = 0$ for $L' < y < L$, the channel voltage at the pinch-off point remains equal to V_{DSAT} .

➤ $V_C(y = L') = V_{DSAT}$ (40)

- The gradual channel approximation
- is valid in this region. thus, the
- channel current

➤
$$I_{D(sat)} = \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L'} (V_{GS} - V_{T0})^2 \text{(41)}$$

Rohit Lorenzo, VIT-AP



Channel length modulation in an n-channel MOSFET operation in saturation

Channel length modulation

- Note that this current equation corresponds to a MOSFET with effective channel length L' . We can rewrite the saturation current as follows

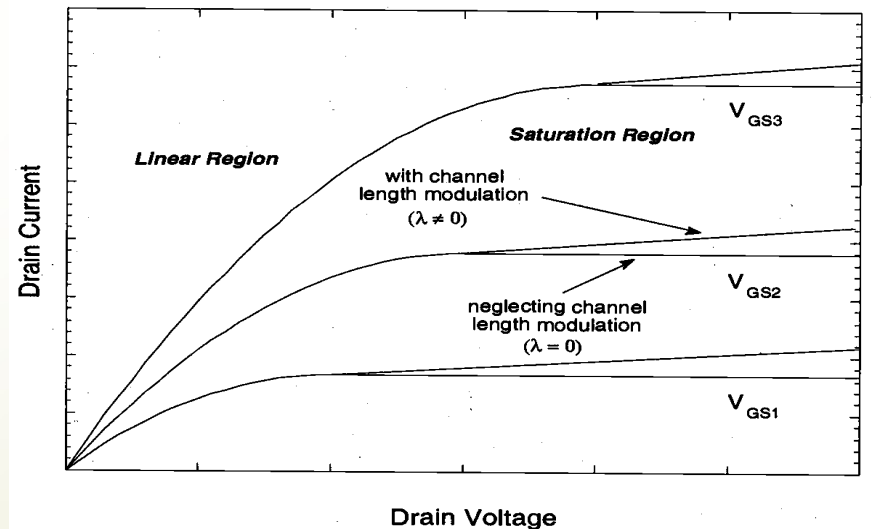
$$I_{D(sat)} = \left(\frac{1}{1 - \frac{\Delta L}{L}} \right) \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L} (V_{GS} - V_{T0})^2 \dots\dots\dots(42)$$

- To simplify the analysis, we will use the following empirical relation

$$1 - \frac{\Delta L}{L} \approx 1 - \lambda V_{DS}$$

- $$I_{D(sat)} = \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L} (V_{GS} - V_{T0})^2 (1 + \lambda V_{DS}) \dots(43)$$

- The slope of the current voltage curve in the saturation region is determined by the channel length modulation
- coefficient λ



Current-voltage characteristics of an n-channel MOS transistor, including the channel length modulation effect.

Body Effect (Substrate bias effect)

- Until now, we have considered a transistor to be a three-terminal device with gate, source, and drain. However, the body is an implicit fourth terminal.
- When a voltage V_{SB} is applied between the source and body, it change the amount of charge required to invert the channel, hence, it alter the threshold voltage.

$$V_T = V_{T0} + \gamma(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|})$$

where V_{T0} is the threshold voltage when the source is at the body potential, the substrate-bias effect can significantly change the value of the threshold voltage.

we finally arrive at a complete first-order characterization of the drain (channel) current as a function of the terminal voltages.

$$I_D = f(V_{GS}, V_{DS}, V_{BS})$$

MOSFET Scaling

- ▶ The design of high-density chips in MOS VLSI (Very Large Scale Integration) technology requires that the packing density of MOSFETs used in the circuits is as high as possible and, consequently, that the sizes of the transistors are as small as possible.
- ▶ The reduction of the size, i.e., the dimensions of MOSFETs, is commonly referred to as **scaling**.
- ▶ It is expected that the operational characteristics of the MOS transistor will change with the reduction of its dimensions. Also, some physical limitations eventually restrict the extent of scaling.
- ▶ There are two basic types of size-reduction strategies: **full scaling** (also called constant-field scaling) and **constant voltage scaling**.
- ▶ Scaling of MOS transistors is concerned with systematic reduction of overall dimensions of the devices as allowed by the available technology, while preserving the geometric ratios found in the larger devices.

Rohit Lorenzo, VIT-AP

Continue....

- ▶ The proportional scaling of all devices in a circuit would certainly result in a reduction of the total silicon area occupied by the circuit, thereby increasing the overall functional density of the chip. To describe device scaling, we introduce a constant *scaling factor* $S > 1$.

Year	1985	1987	1989	1991	1993	1995	1997	1999
Feature size (μm)	2.5	1.7	1.2	1.0	0.8	0.5	0.35	0.25

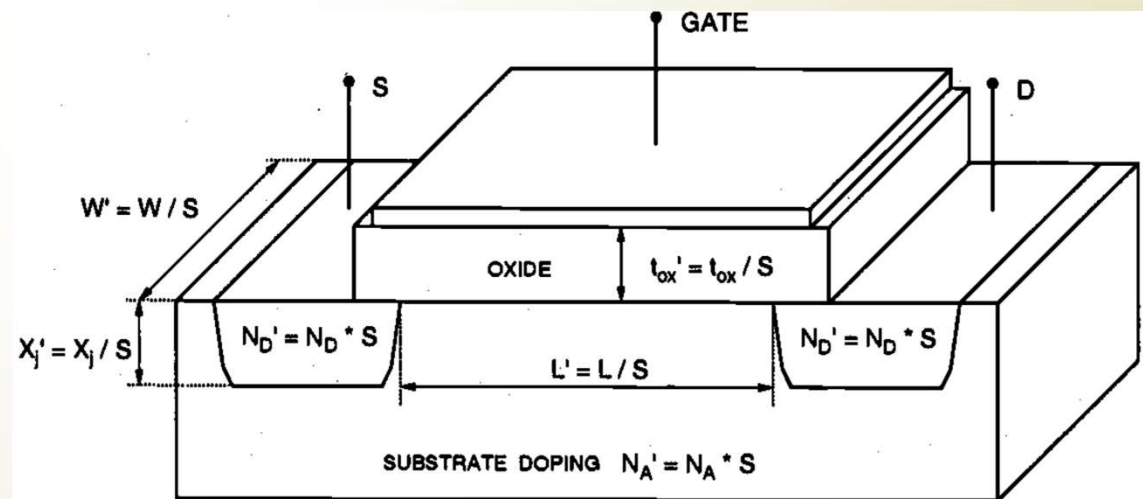


Figure Scaling of a typical MOSFET by a scaling factor of S .

Full Scaling (Constant-Field Scaling)

- This scaling option attempts to preserve the magnitude of internal electric fields in the MOSFET, while the dimensions are scaled down by a factor of S .
- To achieve this goal, all potentials must be scaled down proportionally, by the same scaling factor.
- In order to maintain electric fields charge densities must be *increased* by a factor of S in order to maintain the field conditions.
- Now consider the influence of full scaling described here upon the current-voltage characteristics of the MOS transistor.
- It will be assumed that the surface mobility μ_n is not significantly affected by the scaled doping density.

Rohit Lorenzo, VIT-AP

Quantity	Before Scaling	After Scaling
Channel length	L	$L' = L / S$
Channel width	W	$W' = W / S$
Gate oxide thickness	t_{ox}	$t_{ox}' = t_{ox} / S$
Junction depth	x_j	$x_j' = x_j / S$
Power supply voltage	V_{DD}	$V_{DD}' = V_{DD} / S$
Threshold voltage	V_{T0}	$V_{T0}' = V_{T0} / S$
Doping densities	N_A	$N_A' = S \cdot N_A$
	N_D	$N_D' = S \cdot N_D$

Continue.....

- ▶ The gate oxide capacitance per unit area, on the other hand, is changed as follows.

$$C_{ox}' = \frac{\epsilon_{ox}}{t_{ox}'} = S \cdot \frac{\epsilon_{ox}}{t_{ox}} = S \cdot C_{ox}$$

- ▶ The aspect ratio **W/L** of the MOSFET will remain unchanged under scaling. Consequently, the trans conductance parameter **k_n** will also be scaled by a factor of **S**.
- ▶ The linear-mode drain current of the scaled MOSFET can now be found as

$$\begin{aligned} I_D'(lin) &= \frac{k_n'}{2} \cdot [2 \cdot (V_{GS}' - V_T') \cdot V_{DS}' - V_{DS}'^2] \\ &= \frac{S \cdot k_n}{2} \cdot \frac{1}{S^2} \cdot [2 \cdot (V_{GS} - V_T) \cdot V_{DS} - V_{DS}^2] = \frac{I_D(lin)}{S} \end{aligned}$$

Continue.....

- Similarly, the saturation-mode drain current is also reduced by the same scaling factor.

$$I_D'(sat) = \frac{k_n'}{2} \cdot (V_{GS}' - V_T')^2 = \frac{S \cdot k_n}{2} \cdot \frac{1}{S^2} \cdot (V_{GS} - V_T)^2 = \frac{I_D(sat)}{S}$$

- the instantaneous power dissipated by the device (before scaling) can be found as:

$$P = I_D \cdot V_{DS}$$

- the power dissipation of the transistor will be reduced by the factor S^2

$$P' = I_D' \cdot V_{DS}' = \frac{1}{S^2} \cdot I_D \cdot V_{DS} = \frac{P}{S^2}$$

- This significant reduction of the power dissipation is one of the most attractive features of full scaling.

Rohit Lorenzo, VIT-AP

Continue.....

- the device area reduction by S^2
- In addition, the proportional reduction of all dimensions on-chip will lead to a reduction of various parasitic capacitances and resistances as well, contributing to the overall performance improvement.

Quantity	Before Scaling	After Scaling
Oxide capacitance	C_{ox}	$C_{ox}' = S \cdot C_{ox}$
Drain current	I_D	$I_D' = I_D / S$
Power dissipation	P	$P' = P / S^2$
Power density	$P / Area$	$P' / Area' = P / Area$

Constant-Voltage Scaling

- ▶ The full scaling strategy dictates that the power supply voltage and all terminal voltages be scaled down proportionally with the device dimensions. The scaling of voltages may not be very practical in many cases.
- ▶ In particular, the peripheral and interface circuitry may require certain voltage levels for all input and output voltages which in turn would necessitate multiple power supply voltages and complicated level shifter arrangements. For these reasons, constant-voltage scaling is usually preferred over full scaling.
- ▶ In constant-voltage scaling, all dimensions of the MOSFET are reduced by a factor of S , as in full scaling. The power supply voltage and the terminal voltages, on the other hand, remain unchanged. The doping densities must be increased by a factor of S^2 in order to preserve the charge-field relations.

continue.....

Quantity	Before Scaling	After Scaling
Dimensions	W, L, t_{ox}, x_j	reduced by S ($W' = W / S, \dots$)
Voltages	V_{DD}, V_T	remain unchanged
Doping densities	N_A, N_D	increased by S^2 ($N_A' = S^2 \cdot N_A, \dots$)

- The linear mode drain current of the scaled MOSFET can be written as:

$$\begin{aligned} I_D'(lin) &= \frac{k_n'}{2} \cdot [2 \cdot (V_{GS}' - V_T') \cdot V_{DS}' - V_{DS}'^2] \\ &= \frac{S \cdot k_n}{2} \cdot [2 \cdot (V_{GS} - V_T) \cdot V_{DS} - V_{DS}^2] = S \cdot I_D(lin) \end{aligned}$$

Continue....

- Also, the saturation-mode drain current will be increased by a factor of S after constant voltage scaling.

$$I_D'(sat) = \frac{k_n'}{2} \cdot (V_{GS}' - V_T')^2 = \frac{S \cdot k_n}{2} \cdot (V_{GS} - V_T)^2 = S \cdot I_D(sat)$$

- Next, consider the power dissipation. Since the drain current is increased by a factor of S

$$P' = I_D' \cdot V_{DS}' = (S \cdot I_D) \cdot V_{DS} = S \cdot P$$

- the power density (power dissipation per unit area) is found to increase by a factor of S^3 after constant-voltage scaling.

Continue.....

Quantity	Before Scaling	After Scaling
Oxide capacitance	C_{ox}	$C_{ox}' = S \cdot C_{ox}$
Drain current	I_D	$I_D' = S \cdot I_D$
Power dissipation	P	$P' = S \cdot P$
Power density	$P / Area$	$P' / Area' = S^3 \cdot (P / Area)$

- To summarize, constant-voltage scaling may be preferred over full (constant-field) scaling in many practical cases because of the external voltage-level constraints.
- This large increase in current and power densities may eventually cause serious reliability problems for the scaled transistor, such as electro-migration, hot-carrier degradation, oxide breakdown, and electrical over-stress.

Short Channel Effects

- ▶ A MOSFET can be defined as a short-channel device if the effective channel length L_{eff} approximately equal to the source and drain junction depth x_j .
- ▶ The short-channel effects that arise in this case are attributed to two physical phenomena:
 - ▶ (i) the limitations imposed on electron drift characteristics in the channel
 - ▶ (ii) the modification of the threshold voltage due to the shortening channel length.
- ▶ Limitation in electron drift
- ▶ The lateral electric field E_y along the channel increases, as the effective channel length is decreased. The electron drift velocity V_d tends to saturate at high electric fields. The velocity saturation has very significant implication on I-V characteristics .
- ▶ In short-channel MOS transistors, the carrier velocity in the channel is also a function of the normal (vertical) electric-field component E_x .

Continue.....

- ▶ The vertical field influences the scattering of carriers (collisions suffered by the carriers) in the surface region. The surface electron mobility expressed by following empirical formula –

$$\mu_n(\text{eff}) = \frac{\mu_{no}}{1 + \Theta \cdot E_x} = \frac{\mu_{no}}{1 + \frac{\Theta \cdot \epsilon_{ox}}{t_{ox} \epsilon_{Si}} \cdot (V_{GS} - V_c(y))} \quad \dots (1)$$

- ▶ where μ_{no} is the low-field surface electron mobility and Θ is the empirical factor. The field related mobility reduction can be approximated by

$$\mu_n(\text{eff}) = \frac{\mu_{no}}{1 + \eta \cdot (V_{GS} - V_T)} \quad \dots (2)$$

- ▶ Where η is an empirical coefficient

▶ Modification of threshold voltage

- ▶ We see simplified geometry of the gate-induced bulk depletion region and the pn-junction depletion regions in a short-channel MOS transistor.

Continue.....

- Figure shows bulk depletion region is assumed to be in asymmetric trapezoidal shape. The drain depletion region is larger than the source due to positive V_{DS} . We recognize that a significant portion is actually due to the source and drain junction depletion, rather than the bulk depletion induced by the gate voltage.
- The bulk depletion charge in the short-channel device is smaller than expected, the threshold voltage expression must be modified as -
- $V_{T0}(\text{Short channel}) = V_{T0} - \Delta V_{T0} \dots\dots (3)$
- V_{T0} is at zero bias ΔV_{T0} is the threshold shift reduction due to short channel effect.
- ΔL_S and ΔL_D represent the lateral extent of the depletion regions. the bulk depletion region charge contained within the trapezoidal region is -

$$Q_{B0} = - \left(1 - \frac{\Delta L_S - \Delta L_D}{2L} \right) \sqrt{2q \cdot N_A \epsilon_{Si} |2\phi_F|} \dots(4)$$

Rohit Lorenzo, VIT-AP

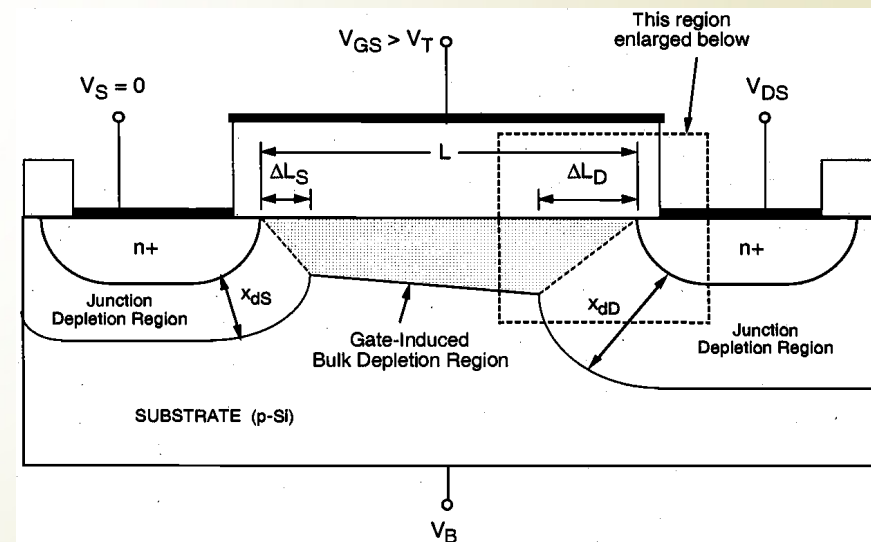


Figure (a) Simplified geometry of the MOSFET channel region, with gate-induced bulk depletion region and the pn-junction depletion regions.

Continue.....

- To calculate ΔL_S and ΔL_D , we will use the simplified geometry as shown in figure. Here, x_{ds} and x_{dD} represent the depth of the pn-junction depletion regions associated with the source and the drain respectively.
- The edges of the source and drain diffusion regions x_j . The vertical extent of the bulk depletion region x_{dm} . The junction depletion region depths can be approximated by-

$$x_{ds} = \sqrt{\frac{2\epsilon_{si}\Phi_o}{q \cdot N_A}} \dots(5)$$

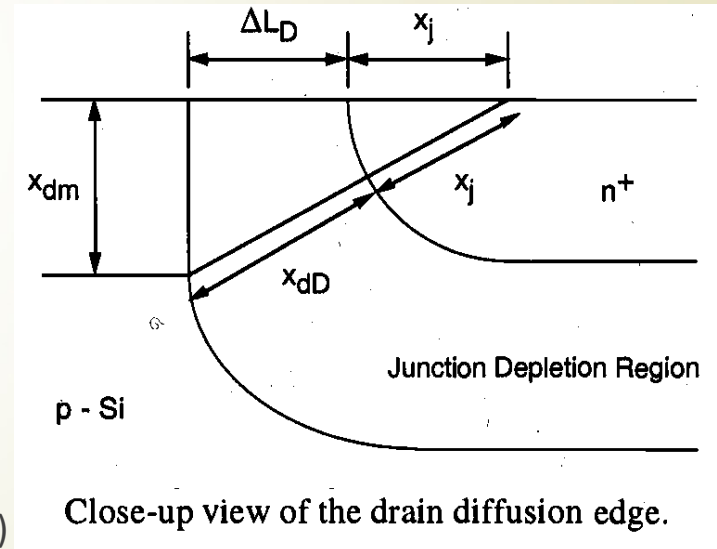
$$x_{dD} = \sqrt{\frac{2\epsilon_{si}(\Phi_o + V_{DS})}{q \cdot N_A}} \dots(6)$$

$$(x_j - x_{dD})^2 = x_{dm}^2 + (x_j - \Delta L_D)^2 \dots(7)$$

$$\Delta L_D^2 + 2x_j \Delta L_D + \Delta L_D^2 + x_{dm}^2 - x_{dD}^2 - 2x_j x_{dD} = 0 \dots(8)$$

Solving ΔL_D

$$\Delta L_D = -x_j + \sqrt{x_j^2 - (x_{dm}^2 - x_{dD}^2) + 2x_j x_{dD}} \cong x_j \cdot \left(\sqrt{1 + \frac{2x_{dD}}{x_j}} - 1 \right) \dots(9)$$



Continue.....

- Similarly length ΔL_S can be found as follows

$$\Delta L_S \cong x_j \cdot \left(\sqrt{1 + \frac{2x_{dS}}{x_j}} - 1 \right) \quad \dots (10)$$

the net amount of threshold voltage reduction ΔV_{T0} due to short-channel effects can be found as

$$\Delta V_{T0} = \frac{1}{C_{ox}} \sqrt{2q \cdot N_A \cdot \epsilon_{si} \cdot |2\phi_F|} \frac{x_j}{2L} \cdot \left[\left(\sqrt{1 + \frac{2x_{dS}}{x_j}} - 1 \right) + \left(\sqrt{1 + \frac{2x_{dD}}{x_j}} - 1 \right) \right] \quad \dots (11)$$

The threshold voltage shift term is proportional to x_j/L

Problem 4

- Consider an n-channel MOS process with the following parameters: substrate doping density $N_A = 10^{16} \text{ cm}^{-3}$, polysilicon gate doping density $N_D = 2 \times 10^{20} \text{ cm}^{-3}$, gate oxide thickness $t_{ox} = 50 \text{ nm}$, oxide-interface fixed charge density $N_{ox} = 4 \times 10^{10} \text{ cm}^{-2}$, source and drain diffusion doping density $N_D = 10^{17} \text{ cm}^{-3}$. In addition, the channel region is implanted with p-type impurities (impurity concentration $N_I = 2 \times 10^{11} \text{ cm}^{-3}$ to adjust the threshold voltage. The junction depth of the source and drain diffusion regions is $X_j = 1.0 \text{ }\mu\text{m}$.

Plot the variation of the zero-bias threshold voltage V_{T0} as a function of the channel length (assume that $V_{DS} = V_{SB} = 0$) also find V_{T0} for $L = 0.7 \text{ }\mu\text{m}$, $V_{DS} = 5\text{V}$, and $V_{SB} = 0$.

Solution

- we have to find the zero-bias threshold voltage which is already calculated in problem no. 1 and was found to be $V_{T0} = 0.40\text{V}$. The additional p-type channel implant will increase the threshold voltage by an amount of qN_I / C_{ox} . we find the long-channel zero-bias threshold voltage.

$$V_{T0} = 0.40 + \frac{q \cdot N_I}{C_{ox}} = 0.40\text{V} + \frac{1.6 \times 10^{-19} \cdot 2 \times 10^{11}}{7.03 \times 10^{-8}} = 0.855\text{V}$$

Continue.....

- ▶ The source and drain junction built-in voltage is

$$\Phi_0 = \frac{KT}{q} \ln \left(\frac{N_D N_A}{n_i^2} \right) = 0.026 \ln \left(\frac{10^{17} \cdot 10^{16}}{2.1 \times 10^{20}} \right) = 0.76V$$

- ▶ The depth of source and drain junction depletion regions

$$x_{dS} = x_{dD} = \sqrt{\frac{2 \cdot \epsilon_{Si}}{q \cdot N_A} \cdot \phi_0} = \sqrt{\frac{2 \cdot 11.7 \cdot 8.85 \times 10^{-14}}{1.6 \times 10^{-19} \cdot 10^{16}} \cdot 0.76} = 31.4 \times 10^{-6} \text{ cm} = 0.314 \mu\text{m}$$

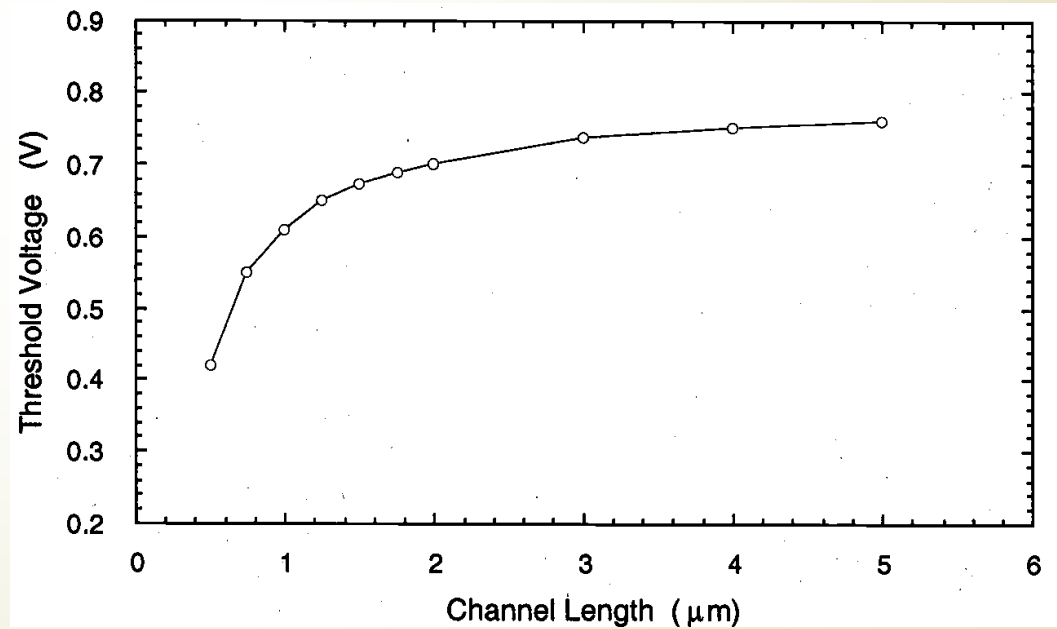
- ▶ the threshold voltage shift V_{T0} , due to short-channel effects can be calculated as a function of the gate (channel) length L .

$$\begin{aligned} \Delta V_{T0} &= \frac{1}{C_{ox}} \cdot \sqrt{2 q \epsilon_{Si} N_A} |2 \phi_F| \cdot \frac{x_j}{2L} \cdot \left[\left(\sqrt{1 + \frac{2x_{dS}}{x_j}} - 1 \right) + \left(\sqrt{1 + \frac{2x_{dD}}{x_j}} - 1 \right) \right] \\ &= \frac{4.82 \times 10^{-8} \text{ C/cm}^2}{7.03 \times 10^{-8} \text{ F/cm}^2} \cdot \frac{1.0 \mu\text{m}}{L} \cdot \left(\sqrt{1 + \frac{2 \cdot 0.314 \mu\text{m}}{1.0 \mu\text{m}}} - 1 \right) \end{aligned}$$

Continue.....

- Finally, the zero-bias threshold voltage is found as

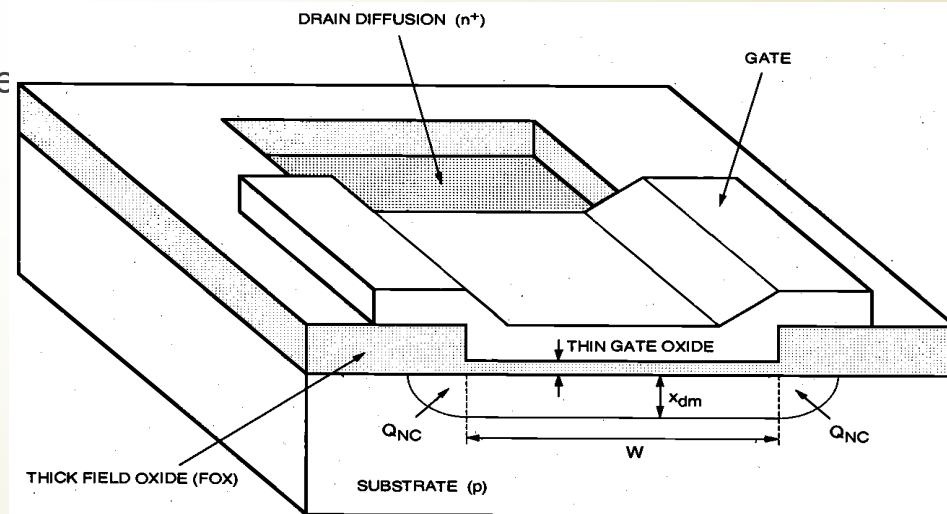
$$V_{T0}(\text{short channel}) = 0.855 \text{ V} - 0.19 \text{ V} \cdot \frac{1}{L[\mu\text{m}]}$$



Narrow Channel Effects

- MOS transistors that have channel widths W on the same order of magnitude as the maximum depletion region thickness x_{dm} are defined as narrow-channel devices.
- The most significant narrow-channel effect is that the actual threshold voltage of such a device is larger than that predicted by the conventional threshold voltage formula.
- A typical cross-sectional view of a narrow-channel device. The oxide thickness in the channel region is t_{ox} , while the regions around the channel are covered by a thick field oxide (**FOX**). Since the gate electrode also overlaps with the field oxide.
- Since the gate electrode also overlaps with the field oxide
- Consequently, the gate voltage must also support this additional depletion charge in order to establish the conducting channel.
- For MOSFETs with small channel widths, however, the actual threshold voltage increases as a result of this extra depletion charge.

Rohit Lorenzo, VIT-AP



Cross-sectional view (across the channel) of a narrow-channel MOSFET. Note that Q_{NC} indicates the extra depletion charge due to narrow-channel effects.

Continue.....

- Finally, the zero-bias threshold voltage is found as

$$V_{T0}(\text{Narrow channel}) = V_{T0} + \Delta V_{T0} \quad \text{.....(1)}$$

- The additional contribution to the threshold voltage due to narrow-channel effects can be modeled as follows:

$$\Delta V_{T0} = \frac{1}{C_{ox}} \cdot \sqrt{2q \cdot N_A \cdot \epsilon_{si} \cdot |2\phi_F|} \frac{K \cdot x_{dm}}{W} \quad \text{.....(2)}$$

where K is an empirical parameter depending on the shape of the fringe depletion region. Depletion region edges are modeled by quarter-circular arcs.

- In all cases, we recognize that the additional contribution to V_{T0} is proportional to (x_{dm}/W) .

Other limitations imposed by small geometries

- ▶ The small-geometry device characteristics, may severely restrict the operating conditions of the transistor and imposed limitations upon the practical utility of the device.
- ▶ Accurate identification and characterization of these small-geometry effects are crucial, especially for submicron MOSFETs.
- ▶ **Velocity Saturation**
- ▶ At high field strengths, the carriers fail to follow this linear model. In fact, when the electrical field along the channel reaches a critical value ξ_c . The velocity of the carriers tends to saturate due to scattering effects (collisions suffered by the carriers) .
- ▶ The velocity as a function of the electrical field, plotted in Figure, can be roughly approximated by the following expression

$$\begin{aligned} v &= \frac{\mu_n \xi}{1 + \xi / \xi_c} \quad \text{for } \xi \leq \xi_c \\ &= v_{sat} \quad \text{for } \xi \geq \xi_c \end{aligned} \quad \dots\dots\dots(1)$$

Rohit Lorenzo, VIT-AP

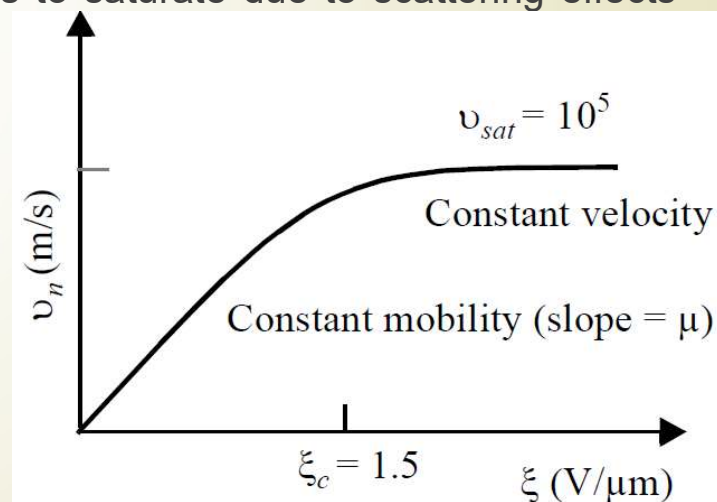


Figure Velocity-saturation effect.

Continue.....

- For a short-channel device and for large enough values of V_{GT} , β is substantially smaller than 1, hence $V_{DSAT} < V_{GS} - V_T$. The device enters saturation before V_{DS} reaches $V_{GS} - V_T$. Short-channel devices therefore experience an extended saturation region, and tend to operate more often in saturation conditions than their long-channel.

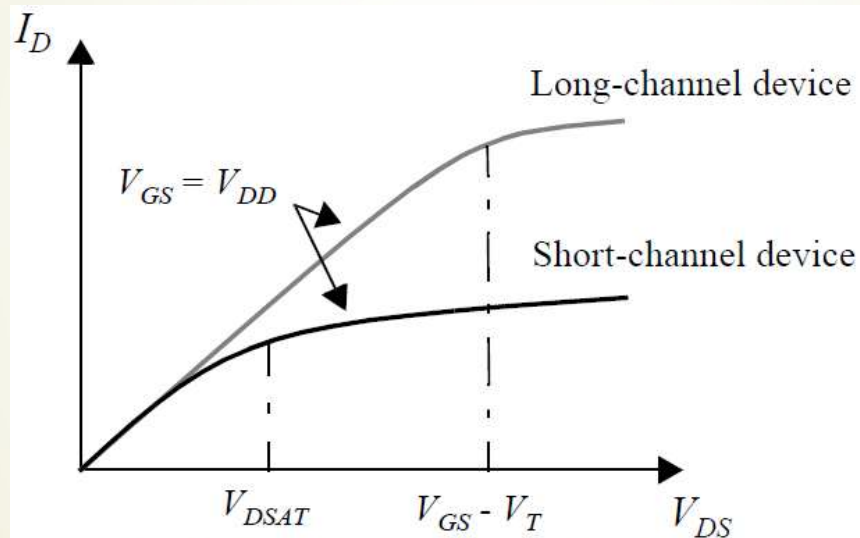


Figure Short-channel devices display an extended saturation region due to velocity saturation.

Drain-Induced Barrier Lowering (DIBL)

- ▶ If the drain voltage is increased, the potential barrier in the channel decreases leading to **drain induced barrier lowering** (DIBL). This is demonstrated by the energy bands along the semiconductor surface
- ▶ For a long channel device, a drain bias can change the effective channel length, but the barrier at the source remains constant. For a short-channel device, this same barrier is no longer fixed. The lowering of the source barrier causes an injection of extra carriers, thereby increasing the current substantially.

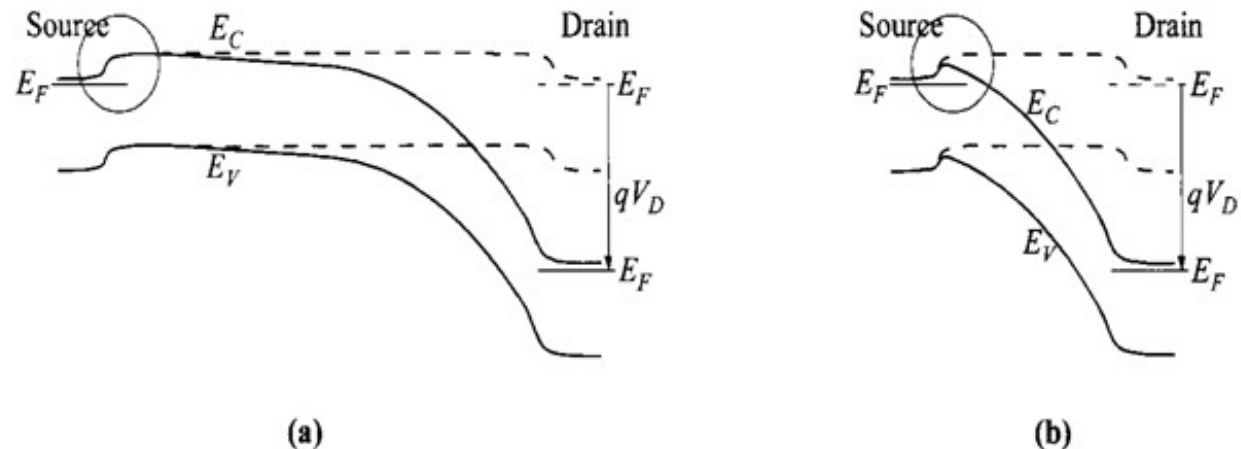


Fig. Energy-band diagram at the semiconductor surface from source to drain, for (a) long-channel and (b) short-channel MOSFETs, showing the DIBL effect in the latter. Dashed lines $V_D = 0$. Solid lines $V_D > 0$.

Subthreshold current

- ▶ The reduction of the potential barrier eventually allows electron flow between the source and the drain, even if the gate-to-source voltage is lower than the threshold voltage. The channel current that flows under these conditions ($V_{GS} < V_{T0}$) is called the subthreshold current. Two-dimensional analysis of the small-geometry MOSFET yields the following approximate expression for the subthreshold current.

$$I_D(\text{subthreshold}) \cong \frac{qD_n W x_c n_0}{L_B} \cdot e^{\frac{q\phi_r}{kT}} \cdot e^{\frac{q}{kT}(A \cdot V_{GS} + B \cdot V_{DS})}$$

- ▶ Here, x_c is the subthreshold channel depth, D_n is the electron diffusion coefficient, L_B is the length of the barrier region in the channel, and ϕ_r is a reference potential.

Oxide breakdown and gate oxide tunneling

- Another limitation on the scaling of t_{ox} is the possibility of oxide breakdown. If the oxide electric field perpendicular to the surface is larger than a certain breakdown field, the silicon-dioxide layer may sustain permanent damage during operation, leading to device failure.
- Reduction in gate oxide thickness results in an increase in electric field across the oxide. This high electric field with low oxide thickness results in tunneling of electrons from substrate to gate and also from gate to substrate through the gate oxide, which results in the gate oxide tunneling.

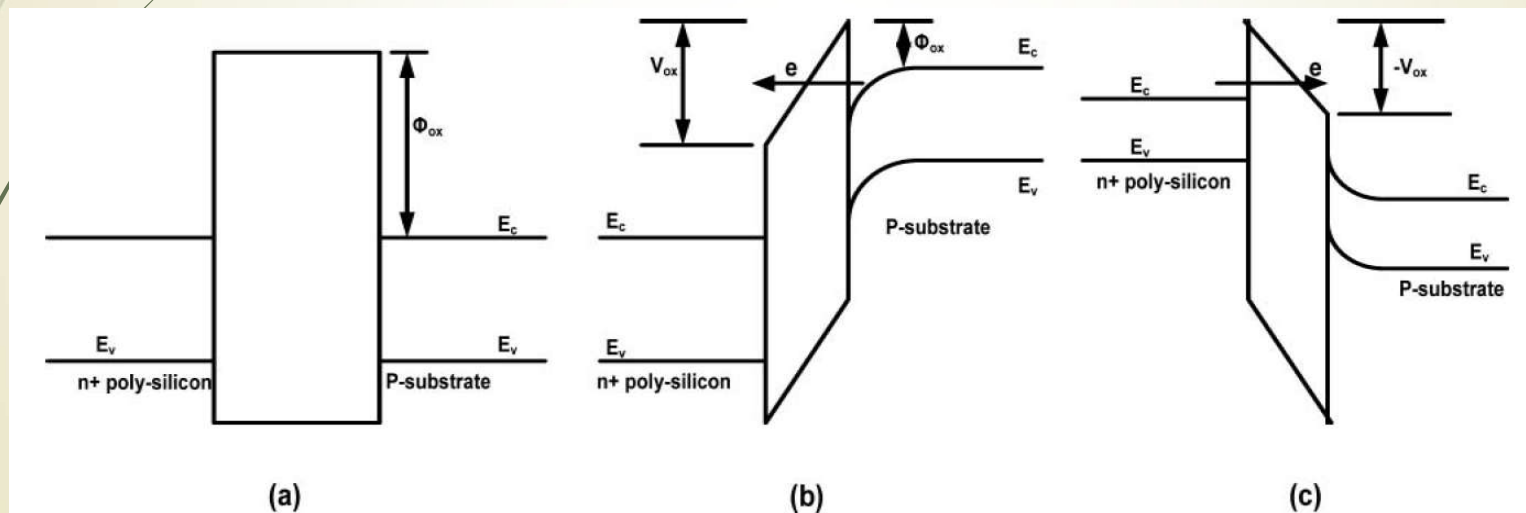


Figure Tunneling of electron through an MOS capacitor. (a) Energy band diagram at flat band condition. (b) Energy band diagram with positive gate bias showing tunneling of electron from substrate to gate. (c) Energy band diagram at negative gate bias showing tunneling of electron from gate to substrate

Hot Carrier Injection

- ▶ The highest field occurs near the drain, and this is the location where most of the anomalous currents originate. The channel carriers (electrons) go through the high-field region, they acquire extra energy from the field without losing it to the lattice. These energetic carriers are called hot **carriers**.
- ▶ Electrons and holes gaining high kinetic energies in the electric field (*hot carriers*) may, however, be injected into the gate oxide, and cause permanent changes in the oxide interface charge distribution, degrading the current-voltage characteristics of the MOSFET.

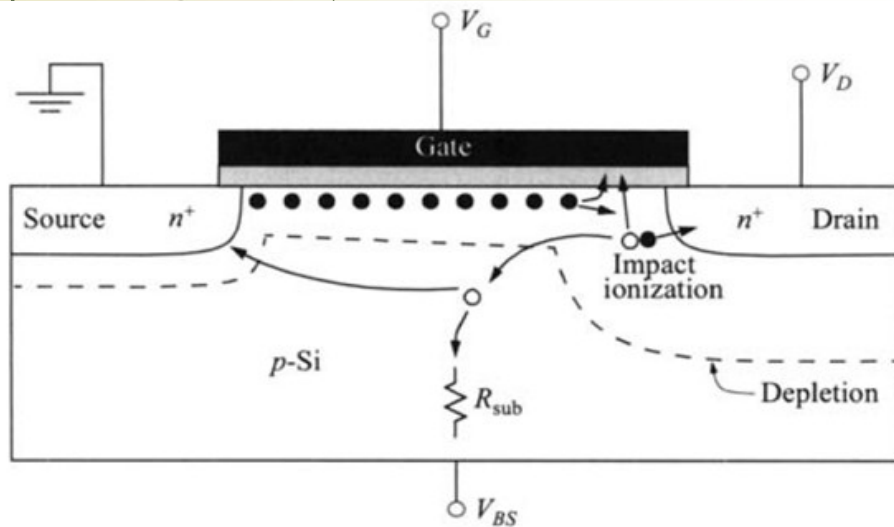


Fig. 1 Current components of a MOSFET under high fields.

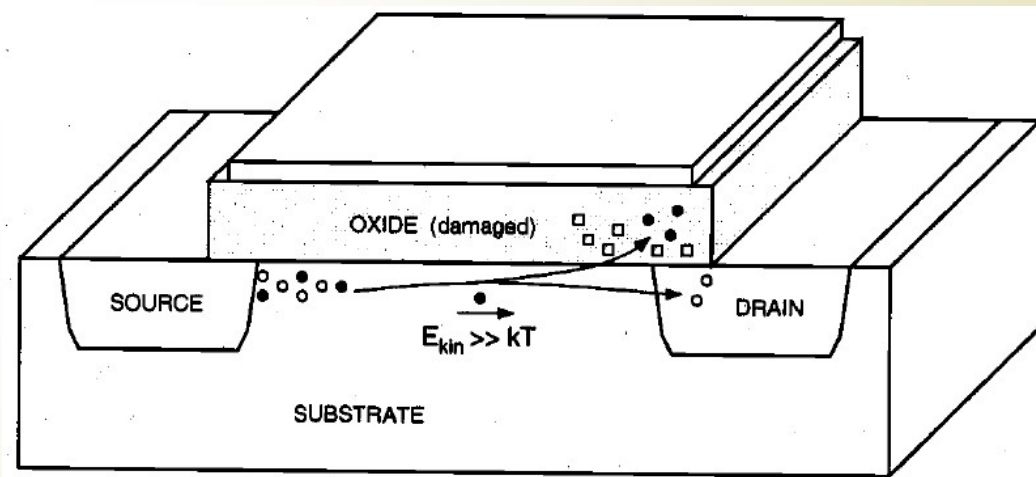
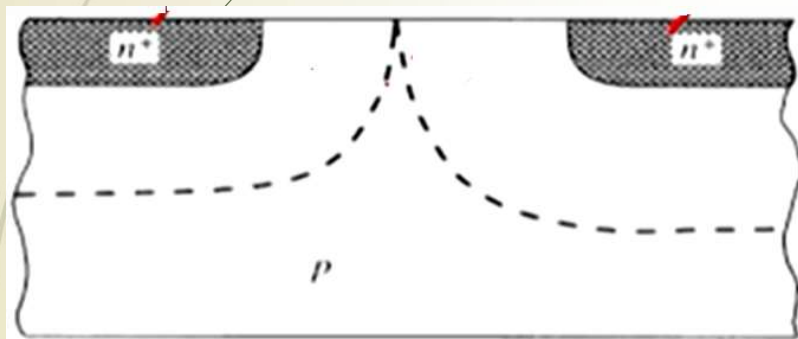


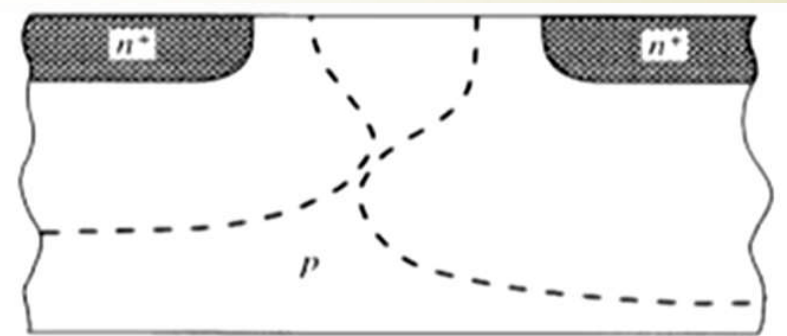
Fig. 2 Hot-carrier injection into the gate oxide and resulting oxide damage.

Punch through

- The drain and source regions are very much close to each other in short-channel devices. The depletion region across the drain-substrate and source-substrate junction extends into the channel region. The separation between the depletion boundaries gradually decreases, as the channel length is reduced. The increase in V_{ds} further pushes the junction near to each other. Thus, the overall effect of short-channel length and reverse biased substrate-drain junction leads to the merging of the two depletion regions, which cause punch through condition. The punch through results the increase in sub-threshold current.



Surface Punch-through

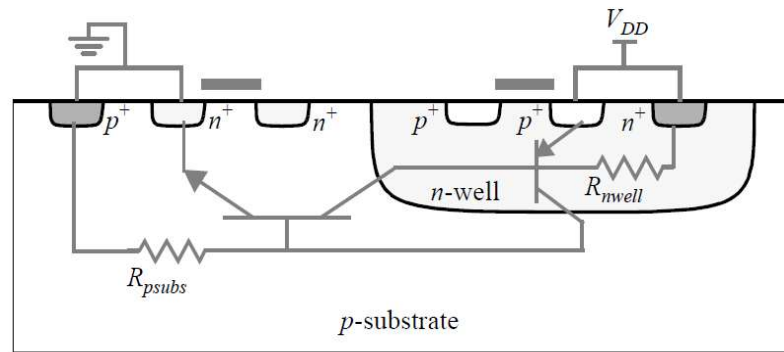


Bulk Punch-through

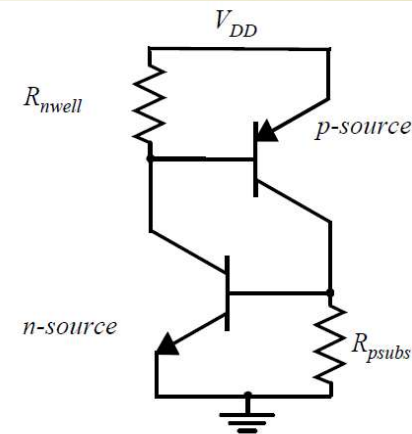
CMOS Latchup

- ▶ The MOS technology contains a number of intrinsic bipolar transistors. These are especially troublesome in CMOS processes, where the combination of wells and substrates results in the formation of parasitic **n-p-n-p** structures. Triggering these thyristor-like devices leads to a shorting of the V_{DD} and V_{SS} lines, usually resulting in a destruction of the chip, or at best a system failure.
- ▶ **Latch-up prevention techniques:**
- ▶ Simply put, latchup prevention/protection includes putting a high resistance in the path so as to limit the current through supply.
- ▶ (1) Surrounding PMOS and NMOS transistors with an insulating oxide layer (trench).
- ▶ (2) Latchup Protection Technology circuitry which shuts off the device when latchup is detected

Rohit Lorenzo, VIT-AP



(a) Origin of latchup



(b) Equivalent circuit

Figure CMOS latchup.

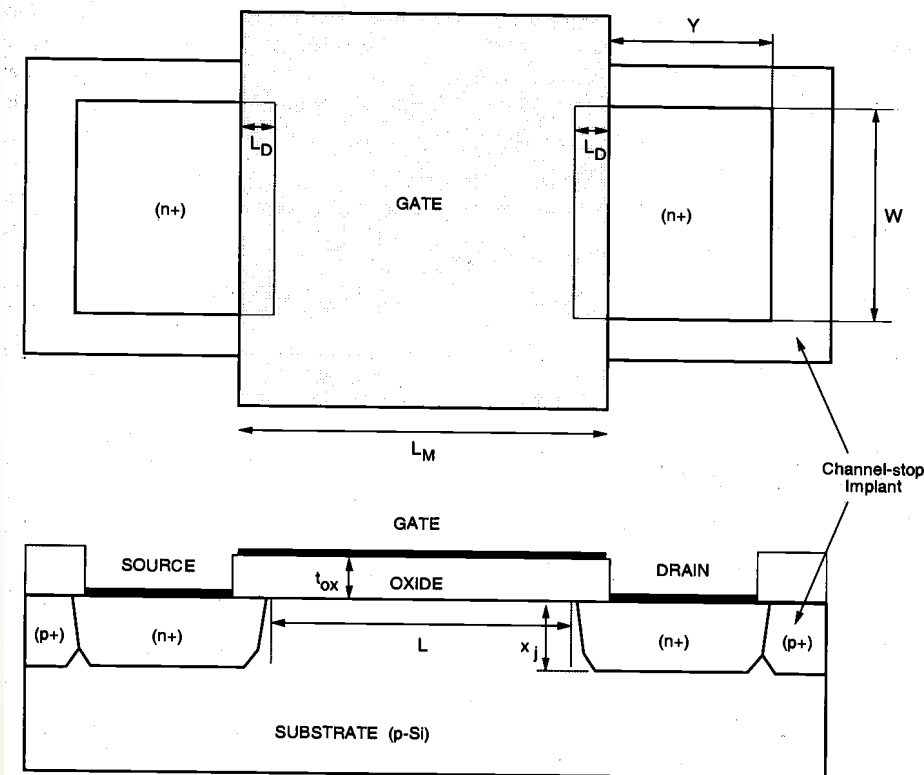
MOSFET Capacitance

- In order to examine the transient (AC) response of MOSFETs and digital circuits consisting of MOSFETs, on the other hand, we have to determine the nature and the amount of parasitic capacitances associated with the MOS transistor.
- In cross-sectional view of figure, the mask length (drawn length) of the gate is indicated by L_M , and the actual channel length is indicated by L .
- The extent of both the gate-source and the gate-drain overlap are L_D ; thus, the channel length is given by

$$L = L_M - 2 \cdot L_D \quad \dots(1)$$

The source and drain overlap region lengths are usually equal to each other. Note that both of these overlap capacitances do not depend on the bias conditions, i.e., they are voltage-independent.

Rohit Lorenzo, VIT-AP

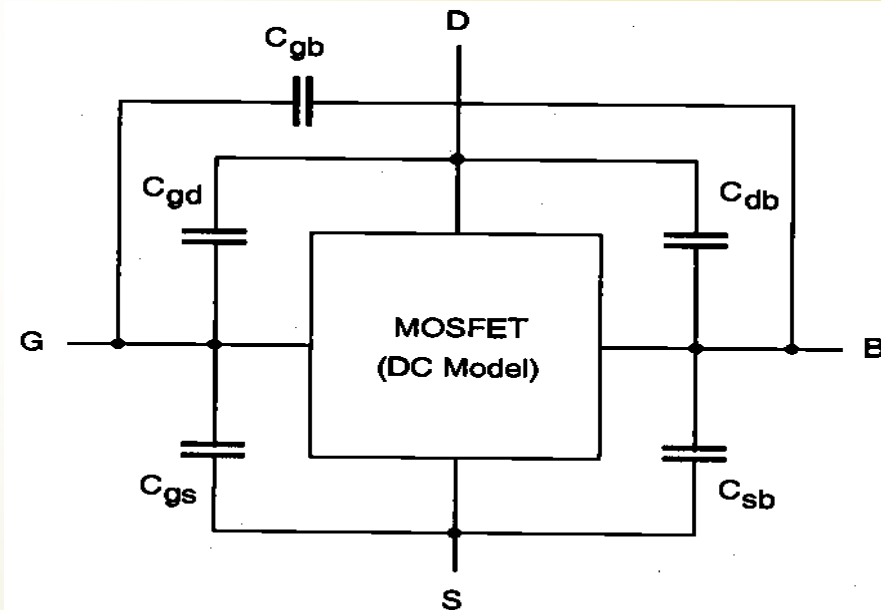


Cross-sectional view and top view (mask view) of a typical n-channel MOSFET

Continue....

- ▶ **P+** doped region, also called the channel-stop implant. this additional p+ region is to prevent the formation of any unwanted (parasitic) channels between two neighboring **n+** diffusion regions. act to electrically isolate neighboring devices built on the same substrate.
- ▶ We will identify the parasitic capacitances associated with this typical MOSFET structure as lumped equivalent capacitances.
- ▶ The parasitic device capacitances can be classified into two major groups: oxide-related capacitances and junction capacitances.

Rohit Lorenzo, VIT-AP



Lumped representation of the parasitic MOSFET capacitances.

Oxide related Capacitance

- ▶ The two overlap capacitances on source and drain as a result of this structural arrangement are called C_{GD} (overlap) and C_{GS} (overlap). Assume same width W for both the diffusion region. The overlap capacitances can be found as

$$\begin{aligned}C_{GS}(\text{overlap}) &= C_{ox}WL_D \\C_{GD}(\text{overlap}) &= C_{ox}WL_D \\C_{ox} &= \epsilon_{ox}/t_{ox}\end{aligned}\quad \text{.....(2)}$$

- ▶ The channel region is connected to the source, the drain, and the substrate, we can identify three capacitances between the gate and these regions, i.e., C_{gs} , C_{gd} and C_{gb} respectively. C_{gs} and C_{gd} is actually gate to channel capacitance.
- ▶ A simplified view of their bias-dependence can be obtained by observing the conditions in the channel region during cut-off, linear, and saturation modes.
- ▶ In cut OFF mode $C_{gs} = C_{gd} = 0$ because of no conducting channel. The gate-to-substrate capacitance can be approximated by
$$C_{gb} = C_{ox} \cdot W \cdot L \quad \text{.....(3)}$$

Continue....

- ▶ In linear-mode operation, conducting inversion layer on the surface effectively shields the substrate from the gate electric field; thus, $C_{gb} = 0$. In this case, the distributed gate-to-channel capacitance may be viewed as being shared equally.

$$C_{gs} \cong C_{gd} \cong \frac{1}{2} \cdot C_{ox} \cdot W \cdot L \quad \dots\dots(4)$$

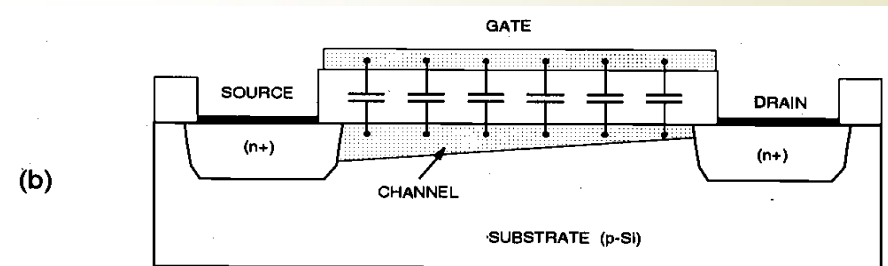
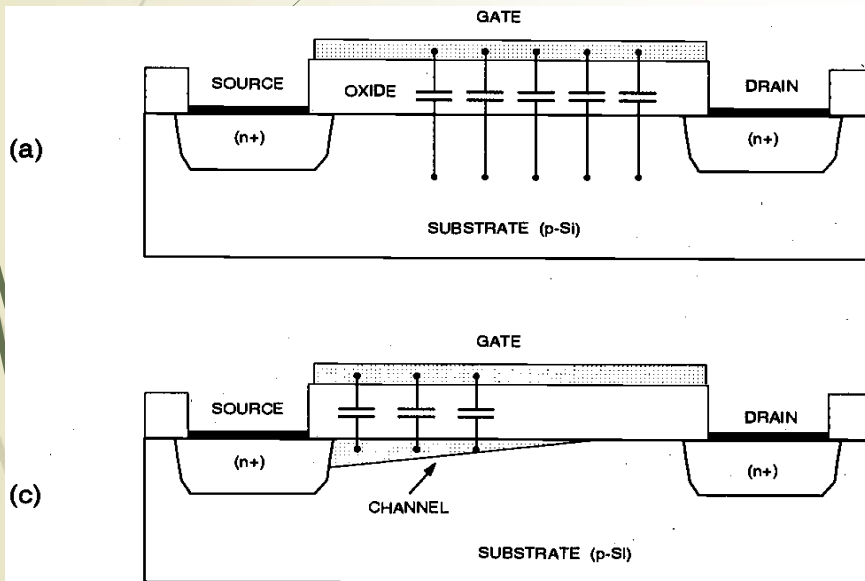


Figure Schematic representation of MOSFET oxide capacitances during (a) cut-off, (b) linear, and (c) saturation modes.

Continue.....

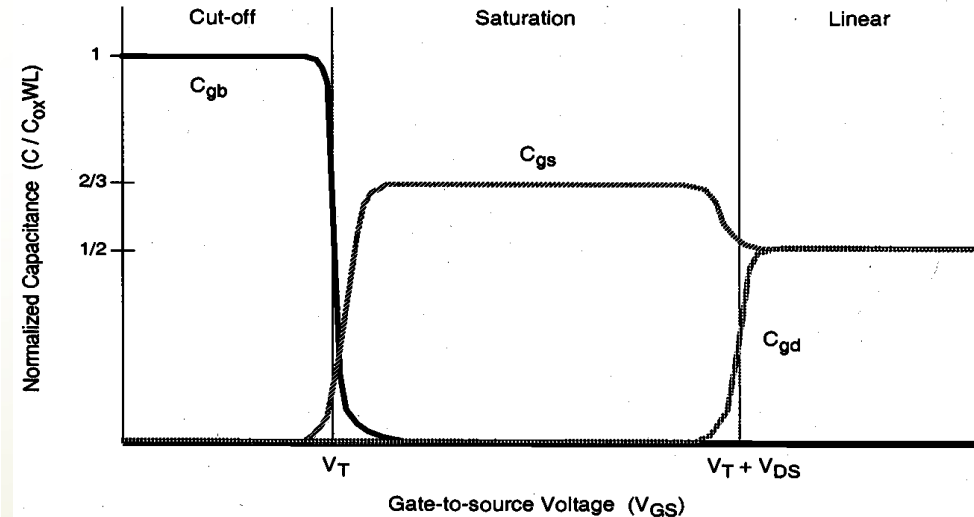
- When the MOSFET is operating in saturation mode, the inversion layer on the surface does not extend to the drain, but it is pinched off. In this case $C_{gd}=0$ and $C_{gb}=0$. Finally, the distributed gate-to-channel capacitance can be approximated as

$$C_{gs} \cong \frac{2}{3} \cdot C_{ox} \cdot W \cdot L \quad \dots\dots\dots(4)$$

- we have to combine the distributed C_{gs} and C_{gd} values found here with the relevant overlap capacitance values, in order to calculate the total capacitance between the external device terminals.

Capacitance	Cut-off	Linear	Saturation
$C_{gb}(\text{total})$	$C_{ox}WL$	0	0
$C_{gd}(\text{total})$	$C_{ox}WL_D$	$\frac{1}{2}C_{ox}WL + C_{ox}WL_D$	$C_{ox}WL_D$
$C_{gs}(\text{total})$	$C_{ox}WL_D$	$\frac{1}{2}C_{ox}WL + C_{ox}WL_D$	$\frac{2}{3}C_{ox}WL + C_{ox}WL_D$

Approximate oxide capacitance values for three operating modes of the MOS transistor.



Variation of the distributed (gate-to-channel) oxide capacitances as functions of gate-to-source voltage V_{GS} .

Junction Capacitances

- we consider the voltage-dependent source-substrate and drain-substrate junction capacitances, C_{sb} and C_{db} , respectively. Both of these capacitances are due to the depletion charge surrounding the respective source or drain diffusion regions embedded in the substrate.
- Figure shows the simplified, partial geometry of a typical n-channel enhancement MOSFET, focusing on the n-type diffusion region within the p-type substrate. the n+ diffusion region forms a number of planar pn-junctions with the surrounding p-type substrate, indicated here with 1 through 5.
- The dimensions of the rectangular box representing the diffusion region are given as W , Y , and X_j .
- we recognize that three of the five planar junctions shown here (2, 3, and 4) are actually surrounded by the p+ channel-stop implant.
- The junction labeled (1) is facing the channel, and the bottom junction (5) is facing the p-type substrate

Rohit Lorenzo, VIT-AP

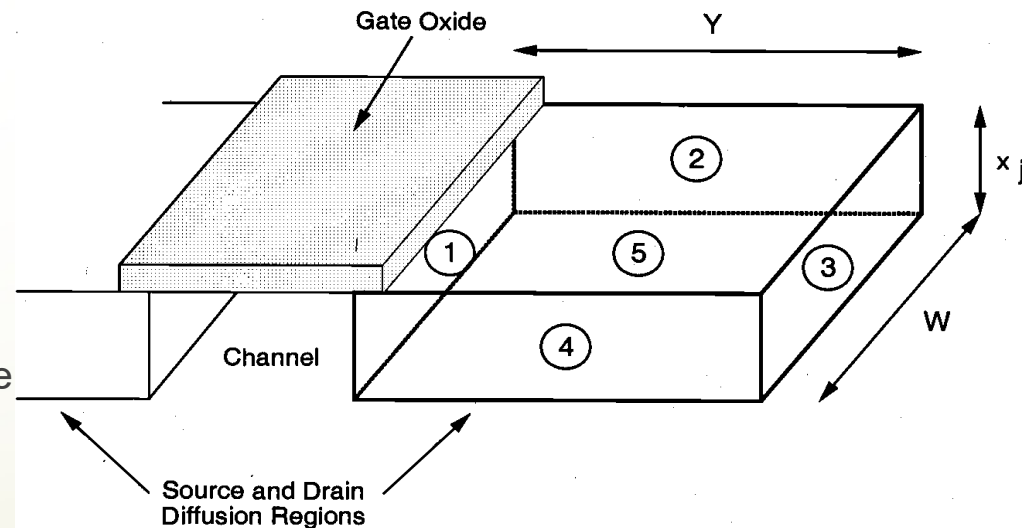


Figure Three-dimensional view of the n⁺ diffusion region within the p-type substrate.

Continue....

- ▶ The junction capacitances associated with these sidewalls will be different from the other junction capacitances.
- ▶ To calculate the depletion capacitance of a reverse-biased abrupt pn-junction, consider first the depletion region thickness, x_d . Assuming that the n-type and p-type doping densities are given by N_D and N_A , respectively, and that the reverse bias voltage is given by V (negative), the depletion region thickness can be found as follows:

$$x_d = \sqrt{\frac{2 \cdot \epsilon_{si}}{q} \frac{N_A + N_D}{N_A \cdot N_D} \cdot (\Phi_0 - V)} \quad \text{.....(5)}$$

- ▶ where the built-in junction potential is calculated as

$$\Phi_0 = \frac{KT}{q} \ln \left(\frac{N_A \cdot N_D}{n_i^2} \right) \quad \text{.....(6)}$$

Rohit Lorenzo, VIT-AP

Junction	Area	Type
1	$W \cdot x_j$	n^+ / p
2	$Y \cdot x_j$	n^+ / p^+
3	$W \cdot x_j$	n^+ / p^+
4	$Y \cdot x_j$	n^+ / p^+
5	$W \cdot Y$	n^+ / p

Types and areas of the
pn-junctions

Continue....

- ▶ The depletion-region charge stored in this area can be written in terms of the depletion region thickness, x_d .

$$Q_j = A \cdot q \cdot \left(\frac{N_A \cdot N_D}{N_A + N_D} \right) \cdot x_d = A \sqrt{2 \cdot \epsilon_{si} \cdot q \left(\frac{N_A + N_D}{N_A \cdot N_D} \right) \cdot (\Phi_o - V)} \quad \text{.....(6)}$$

- ▶ Here, A indicates the junction area. The junction capacitance associated with the depletion region is defined as

$$C_j = \left| \frac{dQ_j}{dV} \right| \quad \text{..... (7)}$$

- ▶ By differentiating with respect to the bias voltage V. we can now obtain the expression for the junction capacitance as follows.

$$C_j(V) = A \sqrt{\frac{\epsilon_{si} \cdot q}{2} \left(\frac{N_A \cdot N_D}{N_A + N_D} \right)} \frac{1}{\sqrt{\Phi_o - V}} \quad \text{.....(8)}$$

- ▶ This expression can be rewritten in a more general form, to account for the junction grading.

$$C_j(V) = \frac{A \cdot C_{jo}}{\left(1 - \frac{V}{\Phi_o} \right)^m} \quad \text{..... (9)}$$

Continue....

- ▶ The parameter **m** is called the grading coefficient. Its value is equal to 1/2 for an abrupt junction profile, and 1/3 for a linearly graded junction profile. The zero-bias junction capacitance per unit area **C_{j0}** is defined as

$$C_{j0} = \sqrt{\frac{\epsilon_{si} \cdot q}{2} \left(\frac{N_A \cdot N_D}{N_A + N_D} \right) \frac{1}{\phi_0}} \quad \text{.....(10)}$$

- ▶ The accurate estimation of the junction capacitances under transient conditions is quite complicated. The problem of estimating capacitance values under changing bias conditions can be simplified, if we calculate a large-signal average (linear) junction capacitance. This equivalent large-signal capacitance can be defined as follows:

$$C_{eq} = \frac{\Delta Q}{\Delta V} = \frac{Q_j(V_2) - Q_j(V_1)}{V_2 - V_1} = \frac{1}{V_2 - V_1} \int_{V_1}^{V_2} C_j(V) dV \quad \text{.....(11)}$$

- ▶ substitute the value from equation (9) in (11).

$$C_{eq} = \frac{A \cdot C_{j0} \cdot \Phi_0}{(V_2 - V_1)(1 - m)} \cdot \left[\left(1 - \frac{V_2}{\phi_0} \right)^{1-m} - \left(1 - \frac{V_1}{\phi_0} \right)^{1-m} \right] \quad \text{.....(12)}$$

Continue....

- For the special case of abrupt pn-junctions, equation

$$C_{eq} = -\frac{2.A.C_{jo}.\Phi_o}{(V_2 - V_1)} \cdot \left[\sqrt{1 - \frac{V_2}{\Phi_o}} - \sqrt{1 - \frac{V_1}{\Phi_o}} \right] \quad \text{.....(13)}$$

- This equation can be rewritten in a simpler form by defining a dimensionless coefficient K_{eq} as follows

$$C_{eq} = A.C_{jo}.K_{eq} \quad \text{.....(14)}$$

$$K_{eq} = -\frac{2.\sqrt{\Phi_o}}{(V_2 - V_1)} (\sqrt{\Phi_o - V_2} - \sqrt{\Phi_o - V_1}) \quad \text{.....(15)}$$

- Where K_{eq} is the voltage equivalence factor

Continue....

- As we discuss earlier the sidewalls of a typical MOSFET source or drain diffusion region are surrounded by a p^+ channel-stop implant. the sidewall zero bias capacitance C_{j0sw} as well as the sidewall voltage equivalence factor $K_{eq}(sw)$ will be different from those of the bottom junction. Assuming that the sidewall doping density is given by $N_A(sw)$, the zero-bias capacitance per unit area can be found as follows:

$$C_{josw} = \sqrt{\frac{\epsilon_{si} \cdot q}{2} \left(\frac{N_A(sw) \cdot N_D}{N_A(sw) + N_D} \right) \frac{1}{\phi_{0sw}}} \quad \text{.....(16)}$$

- where ϕ_{sw} is the built-in potential of the sidewall junctions. Since all sidewalls in a typical diffusion structure have approximately the same depth of x_j we can define a zero bias sidewall junction capacitance per unit length.

$$C_{jsw} = C_{josw} \cdot X_j \quad \text{.....(17)}$$

Continue....

The sidewall voltage equivalence factor $K_{eq}(sw)$ for a voltage swing between V_1 and V_2 is defined as follows

$$K_{eq}(sw) = -\frac{2 \cdot \sqrt{\Phi_{osw}}}{V_2 - V_1} (\sqrt{\Phi_{osw} - V_2} - \sqrt{\Phi_{osw} - V_1}) \quad \dots\dots(18)$$

Combining the equations (16) through (18), the equivalent large-signal junction capacitance $C_{eq}(SW)$ for a sidewall of length (perimeter) P can be calculated as

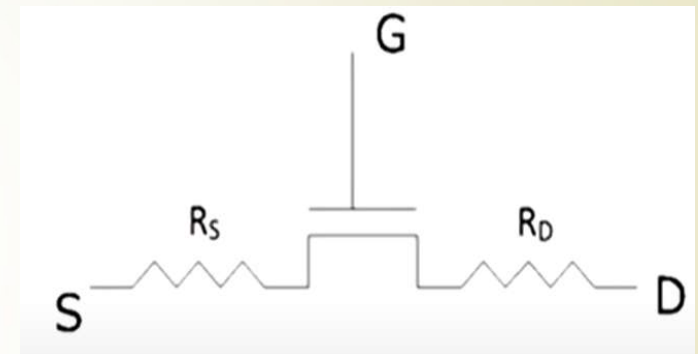
$$C_{eq}(sw) = P \cdot C_{jsw} \cdot K_{eq}(sw) \quad \dots(19)$$

Source – Drain Resistance

- Another parasitic component which effect the performance is the resistance in series with the source and drain.
- The effect is more pronounced in smaller feature size

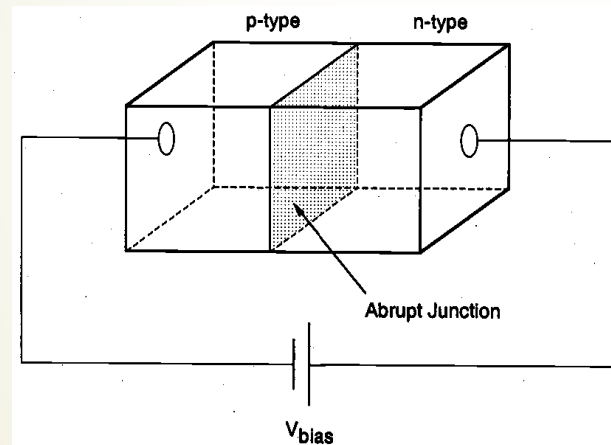
$$R_{S,D} = \frac{L_{S,D}}{W} R_{Sheet} + R_C$$

- Where R_{sheet} and R_C is the sheet and contact resistance respectively



Problem 5

- Consider a simple abrupt pn-junction, which is reverse-biased with a voltage V_{bias} . The doping density of the n-type region is $N_D = 10^{19} \text{ cm}^{-3}$, and the doping density of the p-type region is given as $N_A = 10^{16} \text{ cm}^{-3}$. The junction area is $A = 20 \mu\text{m} \times 20 \mu\text{m}$. consider $V_{\text{bias}} = 5\text{V}$. Calculate (a) the zero-bias junction capacitance per unit area, C_{j0} , for this structure. (b) equivalent capacitance



Continue.....

■ Solution

- First, we will calculate the zero-bias junction capacitance per unit area, C_{j0} , for this structure. The built-in junction potential is found as

$$\Phi_0 = \frac{KT}{q} \ln \left(\frac{N_A \cdot N_D}{n_i^2} \right) = 0.026V \cdot \ln \left(\frac{10^{16} \cdot 10^{19}}{2.1 \times 10^{20}} \right) = 0.88V$$

- we can calculate the zero-bias junction capacitance

$$C_{j0} = \sqrt{\frac{\epsilon_{si} \cdot q}{2} \left(\frac{N_A \cdot N_D}{N_A + N_D} \right) \frac{1}{\Phi_0}} = \sqrt{\frac{11.7 \times 8.85 \times 10^{-14} \times 1.6 \times 10^{-19} C}{2} \left(\frac{10^{16} \cdot 10^{19}}{10^{16} + 10^{19}} \right) \frac{1}{0.88V}} = 3.1 \times 10^{-8} F/cm^2$$

- Next, find the equivalent large-signal junction capacitance assuming that the reverse bias voltage changes from $V_1 = 0$ to $V_2 = -5$ V. The voltage equivalence factor for this transition can be found as follows

$$K_{eq} = -\frac{2 \cdot \sqrt{\Phi_0}}{(V_2 - V_1)} (\sqrt{\Phi_0 - V_2} - \sqrt{\Phi_0 - V_1})$$

Continue.....

$$K_{eq} = -\frac{2 \cdot \sqrt{0.88}}{-5} (\sqrt{0.88 - (-5)} - \sqrt{0.88}) = 0.56$$

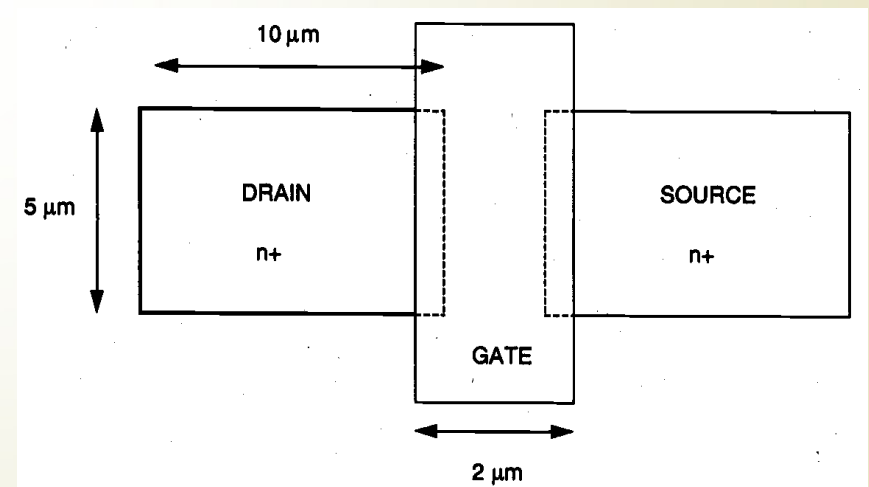
- Then, the average junction capacitance can be found simply by using

$$\begin{aligned} C_{eq} &= A \cdot C_{jo} \cdot K_{eq} = 400 \times 10^{-8} \text{cm}^2 \cdot 3.1 \times 10^{-8} \text{F/cm}^2 \cdot 0.56 \\ &= 69 \text{fF} \end{aligned}$$

Problem 6

- Consider the n-channel enhancement-type MOSFET shown below. The process parameters are given as follows:
- Substrate doping $N_A = 2 \times 10^{15} \text{ cm}^{-3}$
- Source / drain doping $N_D = 10^{19} \text{ cm}^{-3}$
- Sidewall (p+) doping $N_A(\text{sw}) = 4 \times 10^{16} \text{ cm}^{-3}$
- Gate oxide thickness $t_{\text{ox}} = 45 \text{ nm}$
- Junction depth $X_j = 1.0 \text{ } \mu\text{m}$
- Consider reverse bias voltage from $V_{\text{bias}} = 0.5\text{V}$ to 5V
- Calculate (a) the zero bias junction capacitance & Side wall junction capacitance.
- (b) calculate the voltage equivalence factors, for both types of junctions.

Rohit Lorenzo, VIT-AP



Solution

- ▶ We start by calculating the built-in potentials for both types of junctions.

$$\phi_0 = \frac{kT}{q} \cdot \ln\left(\frac{N_A \cdot N_D}{n_i^2}\right) = 0.026 \text{ V} \cdot \ln\left(\frac{2 \times 10^{15} \cdot 10^{19}}{2.1 \times 10^{20}}\right) = 0.837 \text{ V}$$

$$\phi_{0sw} = \frac{kT}{q} \cdot \ln\left(\frac{N_A(sw) \cdot N_D}{n_i^2}\right) = 0.026 \text{ V} \cdot \ln\left(\frac{4 \times 10^{16} \cdot 10^{19}}{2.1 \times 10^{20}}\right) = 0.915 \text{ V}$$

- ▶ Next, we calculate the zero-bias junction capacitances per unit area:

$$\begin{aligned} C_{j0} &= \sqrt{\frac{\epsilon_{Si} \cdot q}{2} \cdot \left(\frac{N_A \cdot N_D}{N_A + N_D}\right) \cdot \frac{1}{\phi_0}} \\ &= \sqrt{\frac{11.7 \cdot 8.85 \times 10^{-14} \text{ F/cm} \cdot 1.6 \times 10^{-19} \text{ C}}{2} \cdot \left(\frac{2 \times 10^{15} \cdot 10^{19}}{2 \times 10^{15} + 10^{19}}\right) \cdot \frac{1}{0.837 \text{ V}}} \\ &= 1.41 \times 10^{-8} \text{ F/cm}^2 \end{aligned}$$

$$\begin{aligned} C_{j0sw} &= \sqrt{\frac{\epsilon_{Si} \cdot q}{2} \cdot \left(\frac{N_A(sw) \cdot N_D}{N_A(sw) + N_D}\right) \cdot \frac{1}{\phi_{0sw}}} \\ &= \sqrt{\frac{11.7 \cdot 8.85 \times 10^{-14} \text{ F/cm} \cdot 1.6 \times 10^{-19} \text{ C}}{2} \cdot \left(\frac{4 \times 10^{16} \cdot 10^{19}}{4 \times 10^{16} + 10^{19}}\right) \cdot \frac{1}{0.915 \text{ V}}} \\ &= 6.01 \times 10^{-8} \text{ F/cm}^2 \end{aligned}$$

Continue.....

- ▶ The zero-bias sidewall junction capacitance per unit length can also be found as follows.

$$C_{jsw} = C_{j0sw} \cdot x_j = 6.01 \times 10^{-8} \text{ F/cm}^2 \cdot 10^{-4} \text{ cm} = 6.01 \text{ pF/cm}$$

- ▶ we must now calculate the voltage equivalence factors, K_{eq} and $K_{eq}(sw)$

$$K_{eq} = -\frac{2\sqrt{0.837}}{-5 - (-0.5)} \cdot (\sqrt{0.837 + 5} - \sqrt{0.837 + 0.5}) = 0.51$$

$$K_{eq}(sw) = -\frac{2\sqrt{0.915}}{-5 - (-0.5)} \cdot (\sqrt{0.915 + 5} - \sqrt{0.915 + 0.5}) = 0.53 \cong K_{eq}$$



Thank You

Rohit Lorenzo, VIT-AP